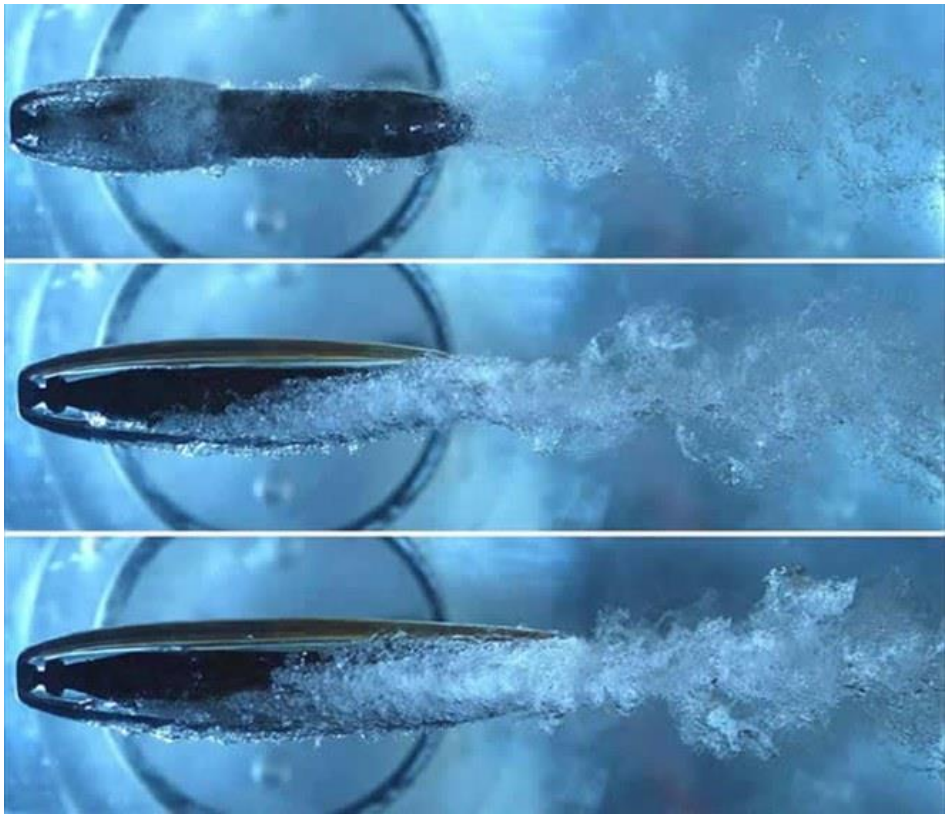
第七章 可压缩粘性流动简介

引 言

可压缩流动的应用







引 言

压缩性的影响： $\frac{\Delta\rho}{\rho} \geq 2\% \sim 5\%$ 热力学过程和动力学过程相耦合

一般认为：理想常比热完全气体绝热流动，不计质量力

$$Re = \frac{UL}{\nu} \sim Ma\frac{aL}{\nu} \gg 1$$

何时需要考虑粘性的影响？

高速流动的传热传质、边界层流动、激波内的流动……

本章主要内容：

了解可压缩粘性流动中的基本概念；

对比不可压和可压缩粘性流动的异同。

基本内容

1. 不可压缩流体温度边界层

2. 可压缩层流运动

3. 可压缩流动的湍流运动

参考教材

《粘性流体力学》，赵学端，廖其奠主编，机械工业出版社

《Viscous Fluid Flow》, Frank M. White, New York : McGraw-Hill, 1991

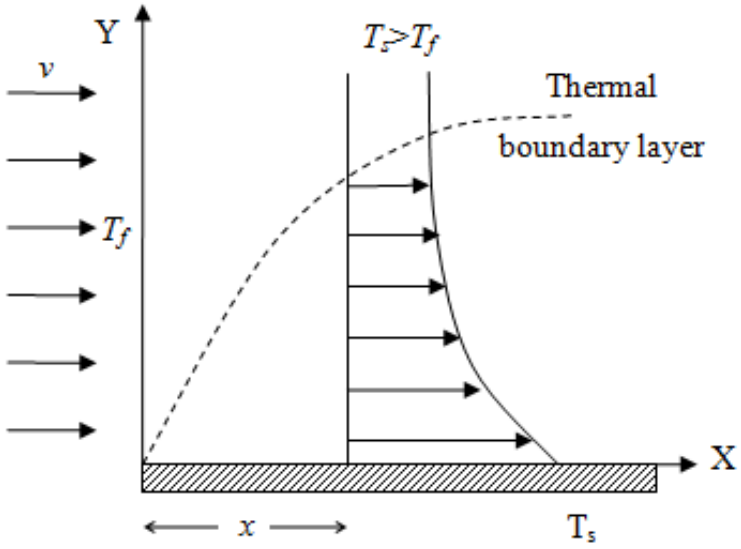
《Boundary-Layer Theory》, Hermann Schlichting, McGraw-Hill, 1979

§7.1不可压缩流体温度边界层

1. 基本概念：

当流体绕流一个吸热或放热的物体时，在其周围就形成了一个温度梯度显著的温度场。如果流体运动速度足够大，温度梯度局限在贴近物面的区域内——温度边界层（热边界层）。

特点：温度沿物面法向梯度很大，热对流和导热具有相同的量级。



温度边界层的名义厚度：

$$T = 0.99T_\infty$$

边界层厚度的估计

$$\nu\frac{\partial^2 U}{\partial y^2}\sim\nu\frac{U}{\delta_u^2} \qquad U\frac{\partial U}{\partial x}\sim\frac{U^2}{L} \qquad \frac{\delta_u}{L}\sim\frac{1}{\sqrt{Re}}$$

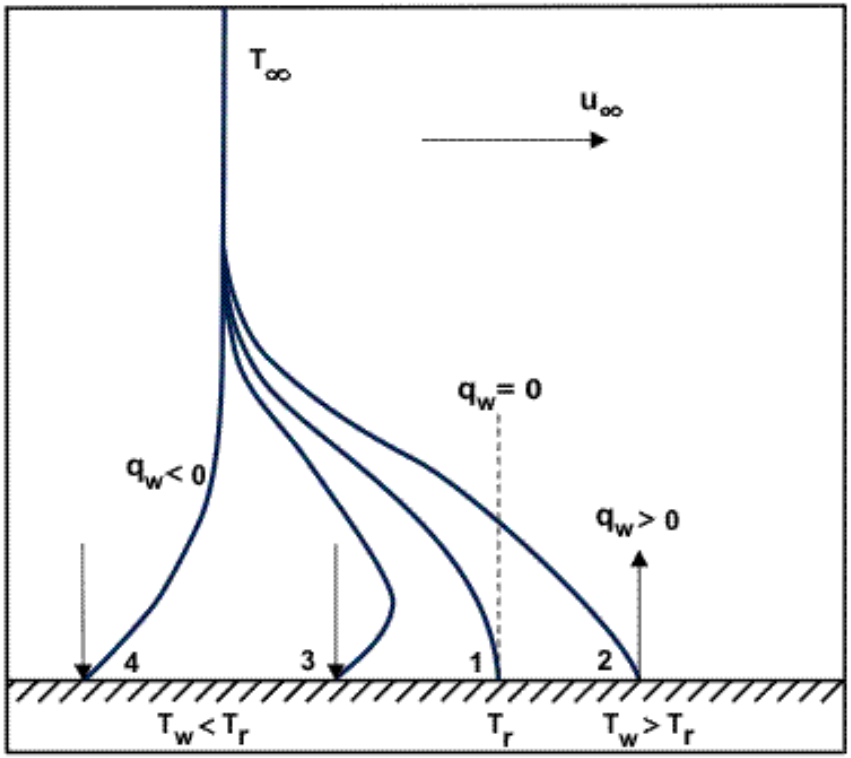
$$\kappa\frac{\partial^2 T}{\partial y^2}\sim\kappa\frac{T}{\delta_T^2} \qquad U\frac{\partial T}{\partial x}\sim\frac{UT}{L} \qquad \frac{\delta_T}{L}\sim\frac{1}{\sqrt{PrRe}}$$

§7.1不可压缩流体温度边界层

壁面热流：$q_w = -\kappa\left(\frac{\partial T}{\partial y}\right)_w$

$$Nu = \frac{q_w L}{\kappa(T_w - T_\infty)} = \frac{\alpha L}{\kappa}$$

$\alpha = \frac{q_w}{T_w - T_\infty}$：放热系数



表征流体与壁面间的热流与流体内传导热之比。

$$St = \frac{q_w}{\rho C_p U_e(T_w - T_\infty)} = \frac{Nu}{RePr}$$

$q_w = 0$：绝热壁面

$q_w < 0$：冷壁面

$q_w > 0$：热壁面

§7.1不可压缩流体温度边界层

2. 能量微分方程：

$$\rho C_p\left(u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}\right)=\kappa\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}\right)+\mu\left[2\left(\frac{\partial u}{\partial x}\right)^2+2\left(\frac{\partial v}{\partial y}\right)^2+\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)^2\right]$$

定义无量纲量 $\theta=\dfrac{T-T_\infty}{T_w-T_\infty}$

$Ec=\dfrac{U_\infty^2}{C_p\left(T_w-T_\infty\right)}$ 埃克尔特数

物理意义？

其他与速度边界层相同

$$u^*\frac{\partial\theta^*}{\partial x^*}+v^*\frac{\partial\theta^*}{\partial y^*}=\frac{1}{PrRe}\left(\frac{\partial^2\theta^*}{\partial x^{*2}}+Re\frac{\partial^2\theta^*}{\partial y^{*2}}\right)+\frac{Ec}{Re}\left[2\left(\frac{\partial u^*}{\partial x^*}\right)^2+2\left(\frac{\partial v^*}{\partial y^*}\right)^2+\left(\sqrt{Re}\frac{\partial u^*}{\partial y^*}+\frac{1}{\sqrt{Re}}\frac{\partial v^*}{\partial x^*}\right)^2\right]$$

由边界层假设，忽略高阶小量：

$$\rho C_p\left(u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}\right)=\kappa\frac{\partial^2 T}{\partial y^2}+\mu\left(\frac{\partial u}{\partial y}\right)^2$$

边界条件：$y=\infty:T=T_\infty$　　$y=0:T=T_w(x)$ or $\dfrac{\partial T}{\partial y}=-\dfrac{1}{\kappa}q_w$

§7.1不可压缩流体温度边界层

对比速度和温度边界层方程：

$$u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=U_e\frac{\mathrm{d}U_e}{\mathrm{d}x}+\nu\frac{\partial^2 u}{\partial y^2}$$

$$\rho C_p\left(u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}\right)=\kappa\frac{\partial^2 T}{\partial y^2}+\mu\left(\frac{\partial u}{\partial y}\right)^2$$

$$\frac{u}{U_\infty}=f_1\left(x^*,y^*\right)=f_1\left(\frac{x}{L},\frac{y}{L}\sqrt{Re}\right)$$

$$\frac{T-T_0}{T-T_\infty}=f_2\left(x^*,y^*\right)=f_2\left(\frac{x}{L},\frac{y}{L}\sqrt{Re},Pr\right)$$

$$\tau_w=\mu\left(\frac{\partial u}{\partial y}\right)_w=\frac{\mu U}{L}\sqrt{Re}F_1\left(\frac{x}{L}\right)$$

$$q_w=-\kappa\left(\frac{\partial T}{\partial y}\right)_w=\frac{\kappa\left(T_w-T_0\right)}{L}\sqrt{Re}F_2\left(\frac{x}{L},Pr\right)$$

$$\rightarrow\quad q_w=\frac{\kappa\left(T_w-T_0\right)}{\mu U}\tau_w F\left(\frac{x}{L},Pr\right)\Leftrightarrow Nu=\frac{1}{2}C_f ReF\left(\frac{x}{L},Pr\right)$$

一般形式的雷诺比拟

若 $Pr=1$ 且压力梯度为0　$Nu=\frac{1}{2}C_f Re$ $\rightarrow$ $St=\frac{Nu}{PrRe}=\frac{1}{2}C_f$

$s=C_f/2St$　雷诺比拟因子　　斯坦顿(Stanton)数

§7.1不可压缩流体温度边界层

3. 相似性解

$$u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=U_e\frac{\mathrm{d}U_e}{\mathrm{d}x}+\nu\frac{\partial^2 u}{\partial y^2}$$

$$\eta=y\sqrt{\frac{U}{\nu x}}\qquad u=Uf'(\eta)\qquad v=\frac{1}{2}\sqrt{\frac{\nu U}{x}}\left[\eta f'(\eta)-f(\eta)\right]$$

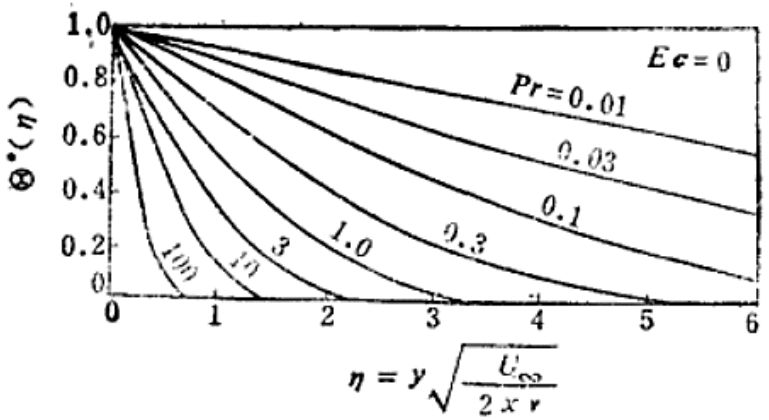
Blasius方程 $2f'''(\eta)+f(\eta)f''(\eta)=0$ 边界条件 $f(0)=f'(0)=0,\quad f'(\infty)=1$

$$\rho C_p\left(u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}\right)=\kappa\frac{\partial^2 T}{\partial y^2}+\mu\left(\frac{\partial u}{\partial y}\right)^2$$

忽略粘性耗散项 $\theta=\dfrac{T-T_0}{T_w-T_0}=g(\eta)$

$$g''+Pr\ fg'=0\qquad g''(0)=1,g(\infty)=0$$

$$g(\eta)=\int_\eta^\infty\left(f''(\eta)\right)^{Pr}d\eta\Big/\int_0^\infty\left(f''(\eta)\right)^{Pr}d\eta$$



§7.1不可压缩流体温度边界层

$$q_w = -\kappa\left(\frac{\partial T}{\partial y}\right)_w = -\frac{\kappa(T_w - T_0)}{L}\sqrt{\frac{U_e}{2\nu x}}g'(0) = \frac{\kappa(T_w - T_0)}{L}\sqrt{\frac{U_e}{2\nu x}}\varphi(Pr) \qquad Nu_x = \sqrt{\frac{Re_x}{2}}\varphi(Pr)$$

根据数值结果近似拟合：

$$Nu = \frac{q_w L}{\kappa(T_w - T_\infty)} = \frac{\alpha L}{\kappa}$$

$$\varphi(Pr) \approx f''(0)Pr^{1/3},\ f''(0) = 0.4696$$

则有：

$$Nu_x \approx 0.332 Pr^{1/3} Re_x^{1/2}$$

根据 $C_f = 0.664 Re_x^{-1/2}$

$$Nu_x = \frac{1}{2} Pr^{1/3} Re_x C_f$$

$$St = \frac{1}{2} Pr^{-2/3} C_f,\ s = \frac{C_f}{2St} = Pr^{2/3}$$

§7.1不可压缩流体温度边界层

考虑粘性耗散项

$$Ec=\frac{U_\infty^2}{C_p\left(T_w-T_\infty\right)}$$

$$g''+Pr\ fg'+PrEcf''^2=0 \qquad g''(0)=1, g(\infty)=0$$

若 $Pr=1$ $\quad g(\eta)=\left(1-\frac{Ec}{2}\right)\left(1-f'(\eta)\right)+\frac{Ec}{2}\left(1-f'(\eta)^2\right)$

$$q_w=\kappa\left(T_w-T_\infty\right)\sqrt{\frac{U_e}{2\nu x}}\left(1-\frac{Ec}{2}\right)f''(0) \qquad St=\frac{f''(0)}{Pr\sqrt{2Re}}\left(1-\frac{Ec}{2}\right)$$

对于绝热平板 $\quad Ec=2 \quad T_{aw}=T_\infty+\frac{U_\infty^2}{2C_p}$

此时**粘性摩擦热产生的绝热壁面温度与势流的绝热滞止温度相等**

速度温度关系式：$\frac{T}{T_\infty}=\frac{T_w}{T_\infty}+\frac{T_{aw}-T_w}{T_\infty}\frac{u}{U_\infty}-\frac{U_\infty^2}{2C_pT_\infty}\left(\frac{u}{U_\infty}\right)^2=\frac{T_{aw}}{T_\infty}-\frac{U_\infty^2}{2C_pT_\infty}\left(\frac{u}{U_\infty}\right)^2$

§7.1不可压缩流体温度边界层

若Pr为任意值：$g''+Pr\ fg'+PrEcf''^2=0$　　$g''(0)=1, g(\infty)=0$

积分两次：

$$g(\eta)=PrEc\int_{\eta}^{\infty}h(\eta,Pr)\left\{\int_{0}^{\eta}\frac{\left(f''(\eta)\right)^2 d\eta}{h(\eta,Pr)}\right\}d\eta-g'(0)\int_{\eta}^{\infty}h(\eta,Pr)d\eta$$

其中：$h(\eta,Pr)=\left[\frac{f''(\eta)}{f''(0)}\right]^{Pr}=\exp\left[-Pr\int_{0}^{\eta}fd\eta\right]$

由$g(0)=1.0$

$$g'(0)=\frac{PrEc\int_{\eta}^{\infty}h(\eta,Pr)\left\{\int_{0}^{\eta}\frac{\left(f''(\eta)\right)^2 d\eta}{h(\eta,Pr)}\right\}d\eta-1}{\int_{\eta}^{\infty}h(\eta,Pr)d\eta}$$

§7.1不可压缩流体温度边界层

绝热壁上，$g'(0)=0$

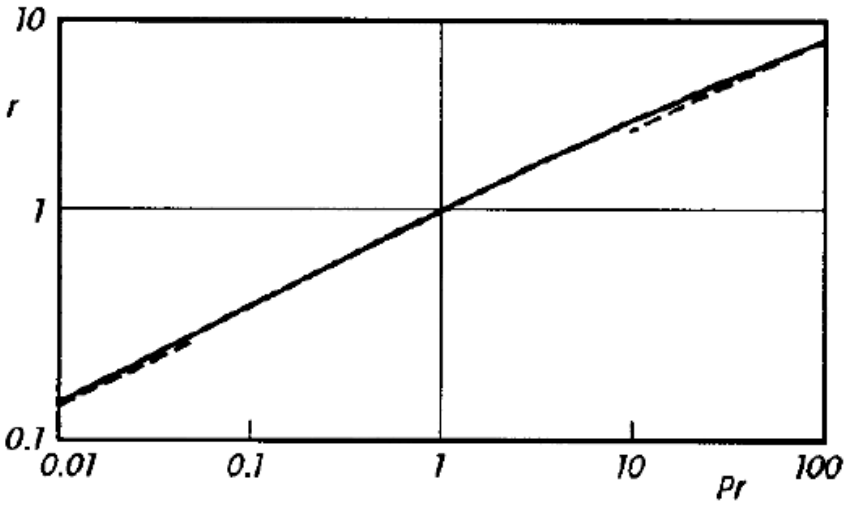
$$Ec=\frac{U_e^2}{C_p\left(T_w-T_\infty\right)}$$

$$Ec_a=\frac{1}{Pr\int_\eta^\infty h(\eta,Pr)\left\{\int_0^\eta\frac{\left(f''(\eta)\right)^2d\eta}{h(\eta,Pr)}\right\}d\eta}\Longrightarrow T_{aw}=T_\infty+r\frac{U_\infty^2}{2C_p}$$

r：恢复因子（**物理意义？与什么有关？**）

$$r=0.9254Pr^{1/2}\left(Pr\to 0\right)$$

$$r=1.9222Pr^{1/3}-1.341\left(Pr\to\infty\right)$$



§7.1不可压缩流体温度边界层

4. 能量积分关系式和边界层厚度的定义

由积分型能量方程有

$$q_w=\frac{d}{dx}\int_0^\infty \rho u\left[(h-h_\infty)-\frac{1}{2}\left(U_\infty^2-u^2\right)\right]dy=\frac{d}{dx}\int_0^\infty \rho u\left(H-H_\infty\right)dy$$ （推导）

焓：$h=e+\frac{p}{\rho}$　　总焓：$H=h+\frac{u^2}{2}$

对于不可压流体，$\rho=const$

热焓厚度

$$q_w=\frac{d}{dx}\left[\rho C_p U_\infty\left(T_w-T_\infty\right)\left(\delta_h-\frac{1}{2}Ec\delta^{**}\right)\right]$$

$$\delta_h=\int_0^\infty \frac{h-h_\infty}{C_p\left(T_w-T_\infty\right)}\frac{u}{U_\infty}dy$$

以上只介绍了绝热平板上的温度边界层流动，其他情况请参考相关教材。

§7.1不可压缩流体温度边界层

5. 湍流温度边界层

平均运动方程

$$\rho C_p \langle u_i \rangle \frac{\partial \langle T \rangle}{\partial x_i} = \frac{\partial}{\partial x_i}\left(\kappa \frac{\partial \langle T \rangle}{\partial x_i} - \rho C_p \langle u_i' T' \rangle \right) + \langle \tau_{ij} \rangle \frac{\partial \langle u_j \rangle}{\partial x_i}$$

雷诺认为，湍流切应力和湍流热流量具有相似的机理，即湍流中的动量传递与热量传递过程是相似的，因而有

$$Pr_t = \frac{-\rho \langle u'v' \rangle}{-\rho C_p \langle u'T' \rangle} \approx 1.0$$

温度壁面律：

$$T^+ = Pru^+ = Pry^+$$

$$T^+ = \frac{1}{\kappa} \ln y^+ + A(Pr) \qquad A(Pr) \approx 12.8 Pr^{0.68} - 7.3$$

§7.2可压缩层流运动

基本方程

$$\frac{\partial \rho}{\partial t}+\mathbf{V}\bullet\nabla\rho=-\rho\nabla\bullet\mathbf{V}$$

$$\frac{\partial \mathbf{V}}{\partial t}+\mathbf{V}\bullet\nabla\mathbf{V}=-\frac{1}{\rho}\nabla p+\frac{1}{\rho}\nabla\left(\lambda\nabla\bullet\mathbf{V}\right)+\frac{1}{\rho}\nabla\bullet\left(2\mu\mathbf{S}\right)$$

$$\frac{\partial e}{\partial t}+\mathbf{V}\bullet\nabla e=-\frac{p}{\rho}\nabla\bullet\mathbf{V}+\frac{1}{\rho}\lambda\left(\nabla\bullet\mathbf{V}\right)^2+\frac{2\mu}{\rho}S_{ij}S_{ij}+\frac{1}{\rho}\nabla\bullet\left(\kappa\nabla T\right)$$

常比热完全气体：

状态方程：$p=\rho RT$　　内能：$e=C_vT$

焓：$h=C_pT$　　熵：$s=C_v\ln(\frac{p}{\rho^\gamma})$

气体常数：$R=C_p-C_v$　　绝热指数：$\gamma=C_p/C_v$

空气：

$R=287\ Nm/kg\cdot K$

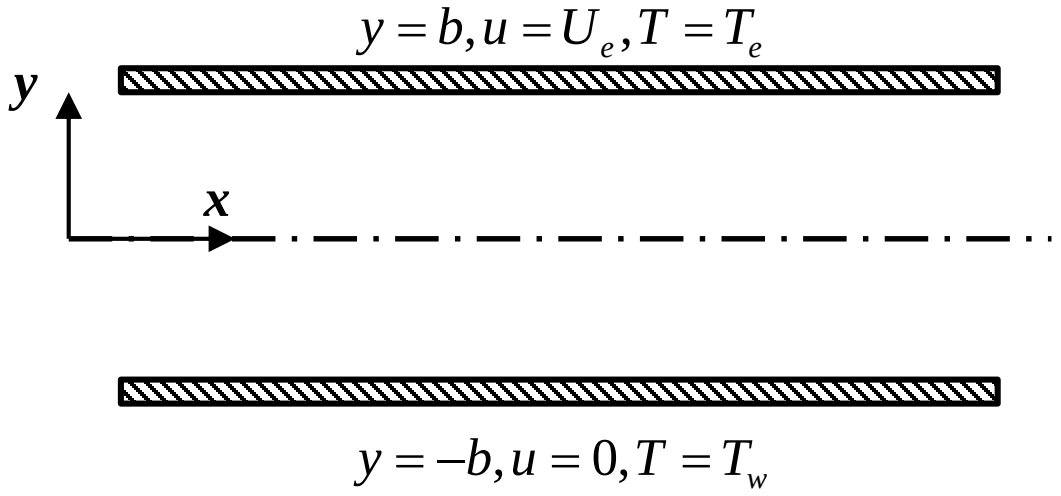
$\gamma=1.4$

§7.2可压缩层流运动

1. 可压缩Couette流

$$\frac{d}{dy}\left(\mu\frac{du}{dy}\right)=0$$

$$\mu\left(\frac{du}{dy}\right)^2+\frac{d}{dy}\left(\kappa\frac{dT}{dy}\right)=0$$



积分上式得 $\tau=\mu\frac{du}{dy}=const$，$\tau u+\kappa\frac{dT}{dy}=const$

下壁面：$u=0,\left(\kappa\frac{dT}{dy}\right)_w=-q_w\Rightarrow\tau u+\kappa\frac{dT}{dy}=-q_w$

$$\mu u\frac{du}{dy}+\kappa\frac{dT}{dy}=-q_w$$ 积分得：$$\frac{u^2}{2}+\int_{T_w}^{T}\frac{\kappa}{\mu}dT=-\frac{q_w}{\tau_w}u$$

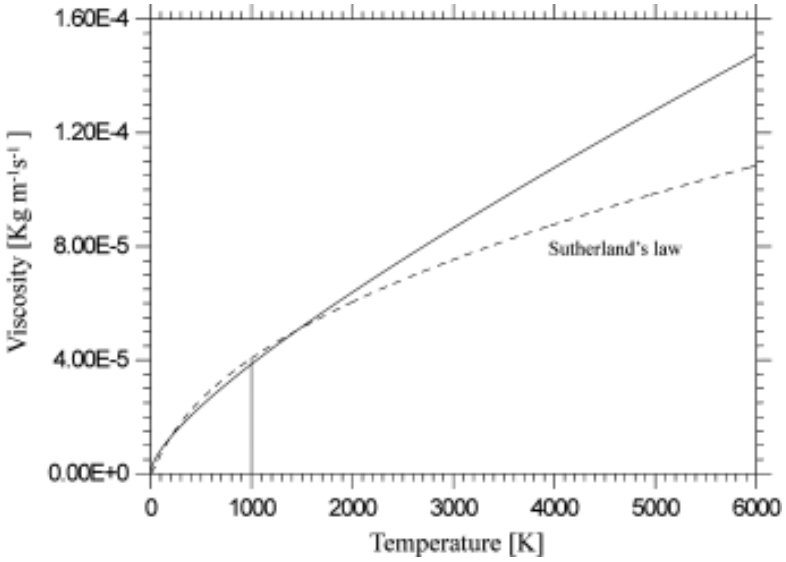
§7.2可压缩层流运动

$$\frac{u^2}{2}+\int_{T_w}^{T}\frac{\kappa}{\mu}dT=-\frac{q_w}{\tau_w}u$$

上壁面：$u=U_e, T=T_e$ $q_w=-\frac{\tau_w}{U_e}\left(\frac{1}{2}U_e^2+\int_{T_w}^{T_e}\frac{\kappa}{\mu}dT\right)$

$$\frac{u^2}{2}+\int_{T_w}^{T}\frac{\kappa}{\mu}dT=\frac{u}{U_e}\left(\frac{1}{2}U_e^2+\int_{T_w}^{T_e}\frac{\kappa}{\mu}dT\right)$$

粘性系数随温度的变化关系：



Sutherland 律 $\mu=\mu_\infty\left(\frac{1+C}{T/T_\infty+C}\right)\left(\frac{T}{T_\infty}\right)^{1.5}$

幂次律 $\mu=\mu_\infty\left(\frac{T}{T_\infty}\right)^{n}$

§7.2可压缩层流运动

$$\frac{\kappa}{\mu}=\frac{C_p}{Pr}=const$$

$$\frac{T}{T_e}=\frac{T_w}{T_e}+\left[\frac{PrEc}{2}+\left(1-\frac{T_w}{T_e}\right)\right]\frac{u}{U_e}-\frac{PrEc}{2}\left(\frac{u}{U_e}\right)^2$$

Crocoo-Busemann速度温度关系式

下壁面为绝热壁：$$\frac{T}{T_e}=\frac{T_{aw}}{T_e}-\frac{PrEc}{2}\left(\frac{u}{U_e}\right)^2$$

恢复因子：$$r=\frac{T_{aw}-T_e}{U_e^2/2C_p}=Pr$$

雷诺比拟因子（下壁面）：$$s=\frac{C_f}{2St}=Pr$$

§7.2可压缩层流运动

2. 可压缩边界层

可压缩边界层流动中既要考虑速度分布，也要考虑温度分布。

速度厚度的估计：$\delta \sim \dfrac{x}{\sqrt{\bar{\rho} U_\infty x / \bar{\mu}}}$

$$\delta_{com} \sim \frac{x}{\sqrt{Re_x}} \left(\frac{\bar{h}}{h_\infty} \right)^{\frac{n+1}{2}} > \left(\frac{x}{\sqrt{Re_x}} \right)_{in} \sim \delta_{in}$$

$$\bar{\rho} \propto 1/\bar{T} \propto 1/\bar{h}$$

$$\bar{\mu} = \mu_\infty \left(\bar{T} / T_\infty \right)^n \propto \mu_\infty \left(\bar{h} / h_\infty \right)^n$$

$$\Rightarrow \bar{\rho} / \bar{\mu} \sim \left(\rho_\infty / \mu_\infty \right) \left(\bar{h} / h_\infty \right)^{\frac{n+1}{2}}$$

马赫数越高，$\bar{h}/h_\infty$越大。即高马赫数可压缩边界层厚度远大于不可压边界层厚度！

温度厚度的估计：

$$\frac{\delta_T}{\delta} \sim \frac{1}{\sqrt{Pr}}$$

§7.2可压缩层流运动

高速气体的可压缩边界层具有以下特点

（1）壁面上气体速度等于0，对边界层流动产生强烈的滞止作用，由于流体的粘性耗散，物面附近的气体温度升高；

（2）可压缩气体速度边界层厚度大于相等雷诺数下不可压边界层

（3）气体绕流的温度边界层厚度大于速度边界层厚度

（4）气体边界层内，流体的密度、粘度和导热系数都是变量，连续方程、动量方程和能量方程耦合，必须联立求解。

§7.2可压缩层流运动

几种厚度的定义	不可压缩边界层	可压缩边界层
位移厚度	$\delta^* = \int_0^{\delta} \left(1 - \frac{u}{U_e}\right) \mathrm{d}y$	$\delta^* = \int_0^{\delta} \left(1 - \frac{\rho u}{\rho_e U_e}\right) \mathrm{d}y$
动量（损失）厚度	$\theta = \int_0^{\delta} \frac{u}{U_e}\left(1 - \frac{u}{U_e}\right) \mathrm{d}y$	$\theta = \int_0^{\delta} \frac{\rho u}{\rho_e U_e}\left(1 - \frac{u}{U_e}\right) \mathrm{d}y$
能量（损失）厚度	$\delta^{**} = \int_0^{\delta} \frac{u}{U_e}\left(1 - \frac{u^2}{U_e^2}\right) \mathrm{d}y$	$\delta^{**} = \int_0^{\delta} \frac{\rho u}{\rho_e U_e}\left(1 - \frac{u^2}{U_e^2}\right) \mathrm{d}y$
热焓厚度	$\delta_h = \int_0^{\infty} \frac{h - h_\infty}{C_p\left(T_w - T_\infty\right)} \frac{u}{U_\infty} dy$	$\delta_h = \int_0^{\infty} \frac{\rho u}{\rho_e U_e}\left(\frac{h}{h_e} - 1\right) dy$

§7.2可压缩层流运动

3. 二维定常可压缩层流边界层的微分方程

$$\frac{\partial \rho u}{\partial x}+\frac{\partial \rho v}{\partial y}=0$$

$$\rho\left(u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}\right)=-\frac{dp}{dx}+\frac{\partial}{\partial y}\left(\mu\frac{\partial u}{\partial y}\right) \qquad p=\rho RT$$

$$\rho\left(u\frac{\partial h}{\partial x}+v\frac{\partial h}{\partial y}\right)=u\frac{\partial p}{\partial x}+\mu\left(\frac{\partial u}{\partial y}\right)^2+\frac{\partial}{\partial y}\left(\frac{\mu}{Pr}\frac{\partial h}{\partial y}\right)$$

$$dh=C_p dT:\rho C_p\left(u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}\right)=u\frac{\partial p}{\partial x}+\mu\left(\frac{\partial u}{\partial y}\right)^2+\frac{\partial}{\partial y}\left(\kappa\frac{\partial T}{\partial y}\right)$$

$$H=h+\frac{u^2}{2}:\rho\left(u\frac{\partial H}{\partial x}+v\frac{\partial H}{\partial y}\right)=\frac{\partial}{\partial y}\left(\left(1-\frac{1}{Pr}\right)\mu u\frac{\partial u}{\partial y}\right)+\frac{\partial}{\partial y}\left(\frac{\mu}{Pr}\frac{\partial H}{\partial y}\right)$$

§7.2可压缩层流运动

下面简单介绍层流边界层的相似性解法

Howarth变换：把密度ρ从原方程中消去，并把方程化成类似不可压缩流的形式。

$$\xi = x \qquad \eta = \int_0^y \frac{\rho}{\rho_\infty} dy$$

$$\rho u = \rho_\infty \frac{\partial \psi}{\partial y}, \rho v = -\rho_\infty \frac{\partial \psi}{\partial x}$$

$$u = \frac{\partial \psi}{\partial \eta}, v = -\frac{\rho_0}{\rho}\left(\frac{\partial \psi}{\partial \xi} + u\frac{\partial \eta}{\partial x}\right)$$

如令：$\tilde{u} = u, \tilde{v} = \frac{\rho}{\rho_0} v + u\frac{\partial \eta}{\partial x}$　　则：$\tilde{u} = \frac{\partial \psi}{\partial \eta}, \tilde{v} = -\frac{\partial \psi}{\partial \xi}$

§7.2可压缩层流运动

经过推导：

$$\frac{\partial \tilde{u}}{\partial \xi}+\frac{\partial \tilde{v}}{\partial \eta}=0$$

$$\tilde{u}\frac{\partial \tilde{u}}{\partial \xi}+\tilde{v}\frac{\partial \tilde{u}}{\partial \eta}=\frac{\rho_\infty}{\rho}\tilde{U}_\infty\frac{d\tilde{U}_\infty}{d\xi}+\nu_\infty\frac{\partial}{\partial \eta}\left(N\frac{\partial \tilde{u}}{\partial \eta}\right)$$

$$\tilde{u}\frac{\partial h}{\partial \xi}+\tilde{v}\frac{\partial h}{\partial \eta}=\frac{\rho_\infty}{\rho}\tilde{U}_\infty\frac{d\tilde{U}_\infty}{d\xi}\tilde{u}+\nu_\infty\frac{\partial}{\partial \eta}\left(\frac{1}{Pr}N\frac{\partial \tilde{u}}{\partial \eta}\right)+\nu_\infty N\left(\frac{\partial \tilde{u}}{\partial \eta}\right)^2$$

$$N=\frac{\rho\mu}{\rho_\infty\mu_\infty}$$

§7.2可压缩层流运动

Illingworth变换：把粘性影响考虑在ξ中，密度影响考虑在η中

$$\xi=\int_0^x\frac{\rho_\infty\mu_\infty}{\rho_0\mu_0}\frac{a_\infty}{a_0}dx \qquad \eta=\frac{a_\infty}{a_0}\int_0^y\frac{\rho}{\rho_0}dy$$

$$\rho u=\rho_0\frac{\partial\psi}{\partial y},\rho v=-\rho_0\frac{\partial\psi}{\partial x}$$

$$\Rightarrow u=\frac{\rho_0}{\rho}\frac{\partial\psi}{\partial y}=\frac{a_\infty}{a_0}\frac{\partial\psi}{\partial\eta},v=-\frac{\rho_0}{\rho}\frac{\partial\psi}{\partial x}=-\frac{\rho_0}{\rho}\frac{\rho_\infty\mu_\infty a_\infty}{\rho_0\mu_0 a_0}\frac{\partial\psi}{\partial\xi}-\frac{\rho_0}{\rho}\frac{a_0}{a_\infty}u\frac{\partial\eta}{\partial x}$$

如令：$$\tilde{u}=\frac{a_0}{a}u,\tilde{v}=\frac{\rho_0\mu_0 a_0}{\rho_\infty\mu_\infty a_\infty}\left(\frac{\rho}{\rho_0}v+\frac{a_0}{a_\infty}u\frac{\partial\eta}{\partial x}\right)$$

则：$$\tilde{u}=\frac{\partial\psi}{\partial\eta},\tilde{v}=-\frac{\partial\psi}{\partial\xi}$$

§7.2可压缩层流运动

经过推导：

$$\frac{\partial \tilde{u}}{\partial \xi}+\frac{\partial \tilde{v}}{\partial \eta}=0$$

$$\tilde{u}\frac{\partial \tilde{u}}{\partial \xi}+\tilde{v}\frac{\partial \tilde{u}}{\partial \eta}=\frac{\rho_\infty}{\rho}\frac{1+\frac{\gamma-1}{2}\frac{u^2}{a^2}}{1+\frac{\gamma-1}{2}\frac{u_\infty^2}{a_\infty^2}}\tilde{U}_\infty\frac{d\tilde{U}_\infty}{d\xi}+\nu_0\frac{\partial}{\partial \eta}\left(\frac{\rho\mu}{\rho_\infty\mu_\infty}\frac{\partial \tilde{u}}{\partial \eta}\right)$$

$$\tilde{u}\frac{\partial H}{\partial \xi}+\tilde{v}\frac{\partial H}{\partial \eta}=\nu_0\frac{\partial}{\partial \eta}\left(\frac{1}{Pr}\frac{\rho\mu}{\rho_\infty\mu_\infty}\frac{\partial \tilde{u}}{\partial \eta}\right)+\nu_0\frac{\partial}{\partial \eta}\left(\left(1-\frac{1}{Pr}\right)\frac{\rho\mu}{\rho_\infty\mu_\infty}\tilde{u}\frac{\partial \tilde{u}}{\partial \eta}\right)$$

进一步令：

$$N=\frac{\rho\mu}{\rho_\infty\mu_\infty},\chi=\frac{H}{H_\infty}-1=H^*-1$$

§7.2可压缩层流运动

$$\frac{\partial \tilde{u}}{\partial \xi}+\frac{\partial \tilde{v}}{\partial \eta}=0$$

$$\tilde{u}\frac{\partial \tilde{u}}{\partial \xi}+\tilde{v}\frac{\partial \tilde{u}}{\partial \eta}=(1+\chi)\tilde{U}_{\infty}\frac{d\tilde{U}_{\infty}}{d\xi}+\nu_0\frac{\partial}{\partial \eta}\left(N\frac{\partial \tilde{u}}{\partial \eta}\right)$$

$$\tilde{u}\frac{\partial \chi}{\partial \xi}+\tilde{v}\frac{\partial \chi}{\partial \eta}=\nu_0\left(\frac{\partial}{\partial \eta}\left(\frac{N}{Pr}\frac{\partial \chi}{\partial \eta}\right)+\frac{\frac{\gamma-1}{2}Ma_{\infty}^2}{1+\frac{\gamma-1}{2}Ma_{\infty}^2}\frac{\partial}{\partial \eta}\left(\left(1-\frac{1}{Pr}\right)N\frac{\partial}{\partial \eta}\left(\frac{\tilde{u}^2}{\tilde{U}_{\infty}^2}\right)\right)\right)$$

边界条件：

$$\eta=0:\tilde{u}=\tilde{v}=0,\chi=\chi_{\infty}\left(or\ \frac{\partial \chi}{\partial \eta}=0\right)$$

$$\eta=\infty:\tilde{u}=\tilde{U}_{\infty},\chi=0$$

§7.2可压缩层流运动

4. 零压梯度平板边界层（ Howarth变换）

$$\frac{\partial \tilde{u}}{\partial \xi}+\frac{\partial \tilde{v}}{\partial \eta}=0$$

$$\tilde{u}\frac{\partial \tilde{u}}{\partial \xi}+\tilde{v}\frac{\partial \tilde{u}}{\partial \eta}=\nu_0\frac{\partial}{\partial \eta}\left(N\frac{\partial \tilde{u}}{\partial \eta}\right)$$

$$\tilde{u}\frac{\partial h}{\partial \xi}+\tilde{v}\frac{\partial h}{\partial \eta}=\nu_\infty\frac{\partial}{\partial \eta}\left(\frac{N}{Pr}\frac{\partial h}{\partial \eta}\right)+\nu_\infty N\left(\frac{\partial \tilde{u}}{\partial \eta}\right)^2$$

$$\tilde{u}=u=U_\infty f'(\zeta)$$

$$\tilde{v}=\sqrt{\frac{\nu_\infty U_\infty}{2\xi}}\left[\zeta f'(\zeta)-f(\zeta)\right]$$

$$h=h_\infty h^*(\zeta)$$

$$\zeta=\eta\sqrt{\frac{U_\infty}{2\nu_\infty \xi}}$$

$$\left(Nf''\right)'+ff''=0$$

$$\left(Nh^{*\prime}\right)'+Prfh^{*\prime}+(\gamma-1)Ma_\infty^2 NPrf''^2=0$$

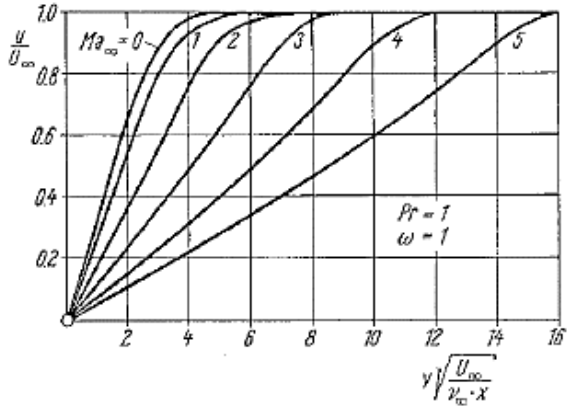
$$B.C.:\ f(0)=0,\ f'(0)=0;$$

$$h^*(0)=h_\infty^*.or.h'^*(0)=0$$

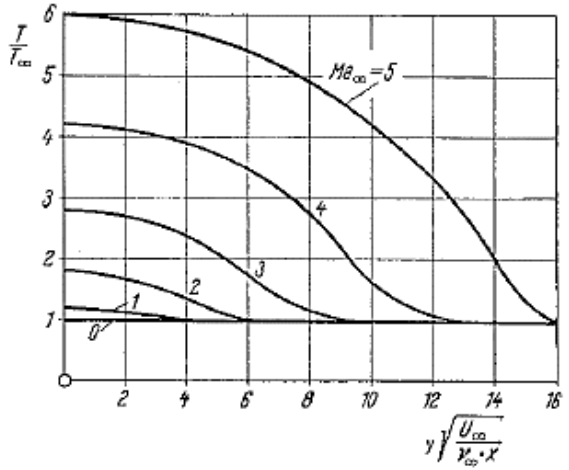
$$f'(\infty)=1,h^*(\infty)=1$$

§7.2可压缩层流运动

绝热壁边界层：Pr=1

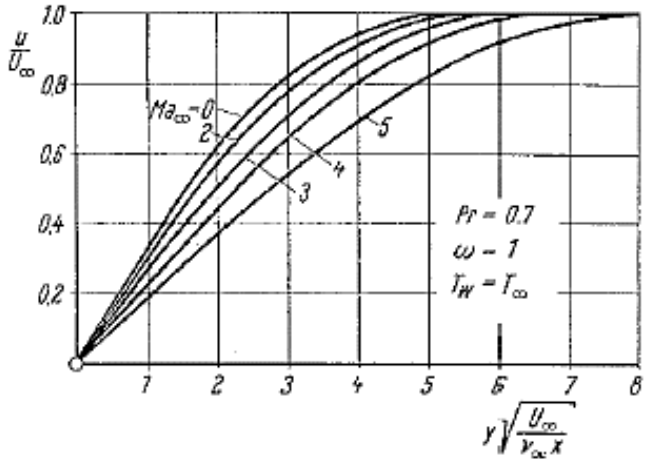


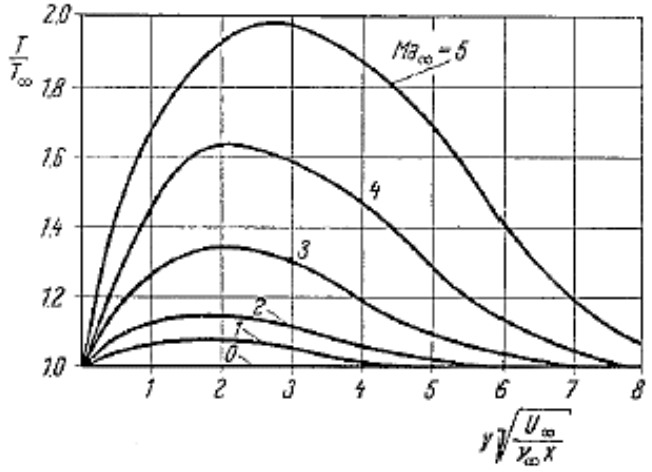
速度分布



温度分布

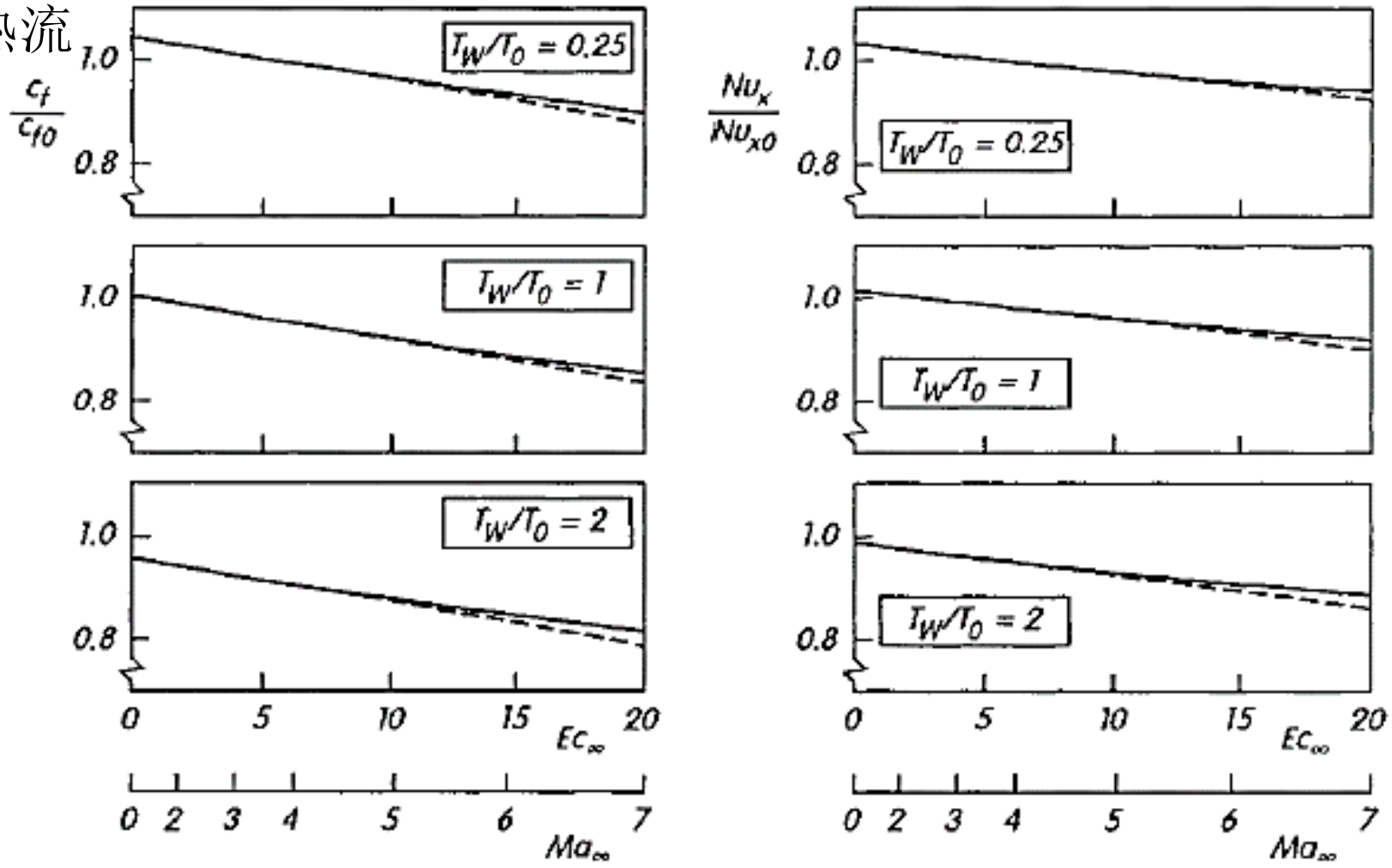
等温壁边界层：Pr=1 $T_w = T_\infty$





§7.2可压缩层流运动

阻力和热流



$$C_{f0} = 0.664 / Re_x^{1/2}, Nu_{x0} = 0.3 / Re_x^{1/2}$$

§7.2可压缩层流运动

5. 有压力梯度的情况

$$N=\frac{\rho\mu}{\rho_\infty\mu_\infty},\chi=\frac{H}{H_\infty}-1=H^*-1$$

存在相似性解的条件：

$$(1)\ Pr=1,\tilde{U}_\infty=const,\chi_w=c\xi^n$$

$$(2)\ Pr=1,N=1,\chi_w=const,\tilde{U}_\infty=c\xi^n$$

$$(3)\ Pr=1,N=1,\chi_w=const,\tilde{U}_\infty=e^{c\xi}$$

$$(4)\ Pr\neq1,\tilde{U}_\infty=const,\chi_w=const$$

提示：采用Illingworth变换

可压缩边界层其他相似性解的例子：

有压力梯度的层流边界层

前驻点附近的流动 $(Nf'')'+ff''-f'^2+h^*=0,\left(N\chi'\right)'+Prf\chi'=0$

§7.2可压缩层流运动

6. Crocco-Busemann速度温度关系式

$$\rho\left(u\frac{\partial H}{\partial x}+v\frac{\partial H}{\partial y}\right)=\frac{\partial}{\partial y}\left(\left(1-\frac{1}{Pr}\right)\mu u\frac{\partial u}{\partial y}\right)+\frac{\partial}{\partial y}\left(\frac{\mu}{Pr}\frac{\partial H}{\partial y}\right)$$

$$Pr=1:\rho\left(u\frac{\partial H}{\partial x}+v\frac{\partial H}{\partial y}\right)=\frac{\partial}{\partial y}\left(\frac{\mu}{Pr}\frac{\partial H}{\partial y}\right)\longrightarrow\left(\frac{\partial h}{\partial y}\right)_w=\left(\frac{\partial H}{\partial y}-u\frac{\partial u}{\partial y}\right)_w=0$$

$H=h+\frac{u^2}{2}=const$ 为满足绝热壁面条件的一个真实解

$$H=h+\frac{u^2}{2}=h_\infty+\frac{U_\infty^2}{2}=H_\infty\Rightarrow T=T_\infty+\frac{1}{2C_p}\left(U_\infty^2-u^2\right)$$

在零压力梯度和 $Pr=1$ 的条件下，能量方程和动量方程形式相同，因此存在下列形式的解

$$H=au+b=h_w+\left(H_\infty-h_w\right)\frac{u}{U_\infty}\longrightarrow T=T_w+\left(T_{aw}-T_w\right)\frac{u}{U_\infty}-\frac{u^2}{2C_p}$$

§7.2可压缩层流运动

在零压力梯度和$Pr \neq 1$的条件下，$T_{aw} = T_\infty + r\dfrac{U_\infty^2}{2C_p}, r = Pr^{1/2}$

$$T = T_w + \left(T_{aw} - T_w\right)\frac{u}{U_\infty} - r\frac{u^2}{2C_p}$$

进而有壁面热流和壁面摩擦之间的关系：

$$q_w = -\left(\kappa\frac{\partial T}{\partial y}\right)_w = \frac{\left(T_w - T_{aw}\right)\kappa_w}{U_\infty \mu_w}\tau_w \Leftrightarrow St = \frac{C_f}{2}$$

§7.2可压缩层流运动

7. 二维定常可压缩层流边界层的积分关系式（自学）

	不可压缩流动	可压缩流动
动量积分	$\frac{c_w}{2}=\frac{\mathrm{d}\theta}{\mathrm{d}x}+\frac{1}{U_e}\frac{\mathrm{d}U_e}{\mathrm{d}x}(2+H)\theta$	$\frac{c_w}{2}=\frac{\mathrm{d}\theta}{\mathrm{d}x}+\frac{1}{U_e}\frac{\mathrm{d}U_e}{\mathrm{d}x}\left(2+H+\frac{U_e}{\rho_e}\frac{d\rho_e}{dU_e}\right)\theta$
动能积分	$\frac{1}{U_e^3}\frac{d}{dx}\left(U_e^3\delta^{**}\right)=\frac{2D}{\rho U_e^3}$	$\frac{1}{\rho_e U_e^3}\frac{d}{dx}\left(\rho_e U_e^3\delta^{**}\right)+\frac{2\delta_h}{U_e}\frac{dU_e}{dx}=\frac{2D}{\rho U_e^3}$
能量积分	$q_w=\frac{d}{dx}\left[\rho C_p U_e\left(T_w-T_e\right)\left(\delta_h-\frac{1}{2}Ec\delta^{**}\right)\right]$	$q_w=\frac{d}{dx}\left[\rho_e U_e\left(h_e\delta_h-\frac{1}{2}U_e^2\delta^{**}\right)\right]$

$$D=\int_0^{\delta}\mu\left(\frac{\partial u}{\partial y}\right)^2 dy$$

§7.2可压缩层流运动

8. 可压缩射流

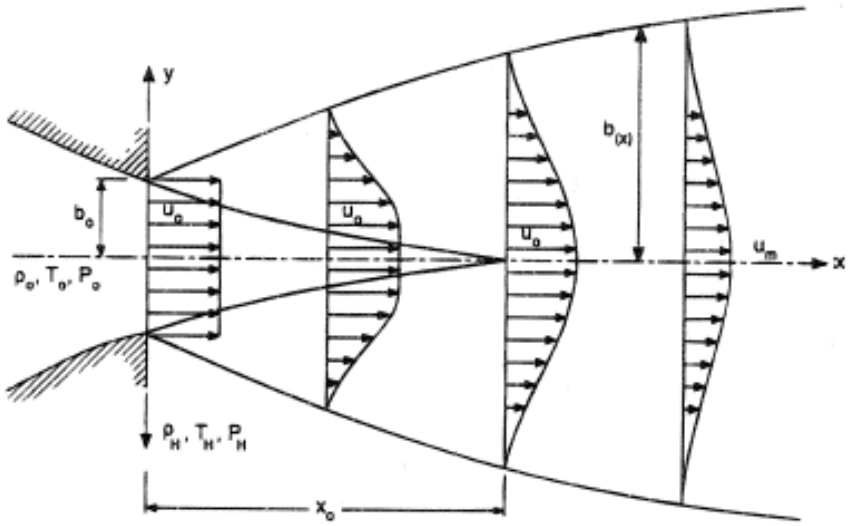
亚音速：与不可压射流类似

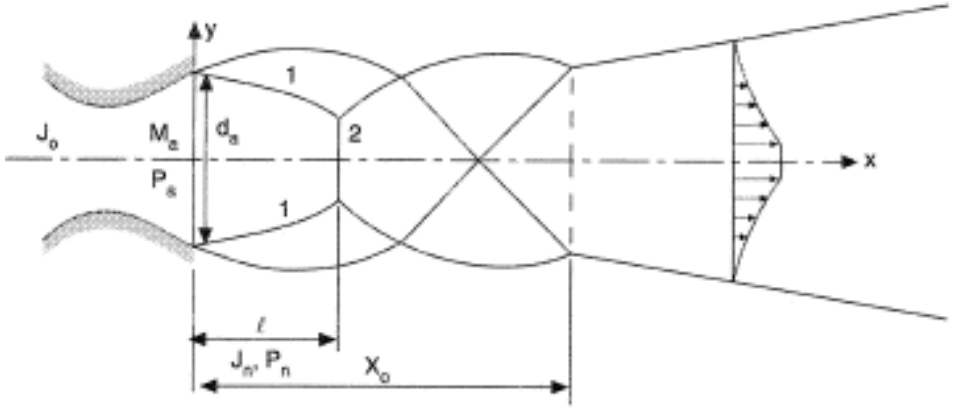
超音速：出口截面流动复杂

出口处可能出现的几种情况：

等熵流、正激波、斜激波……

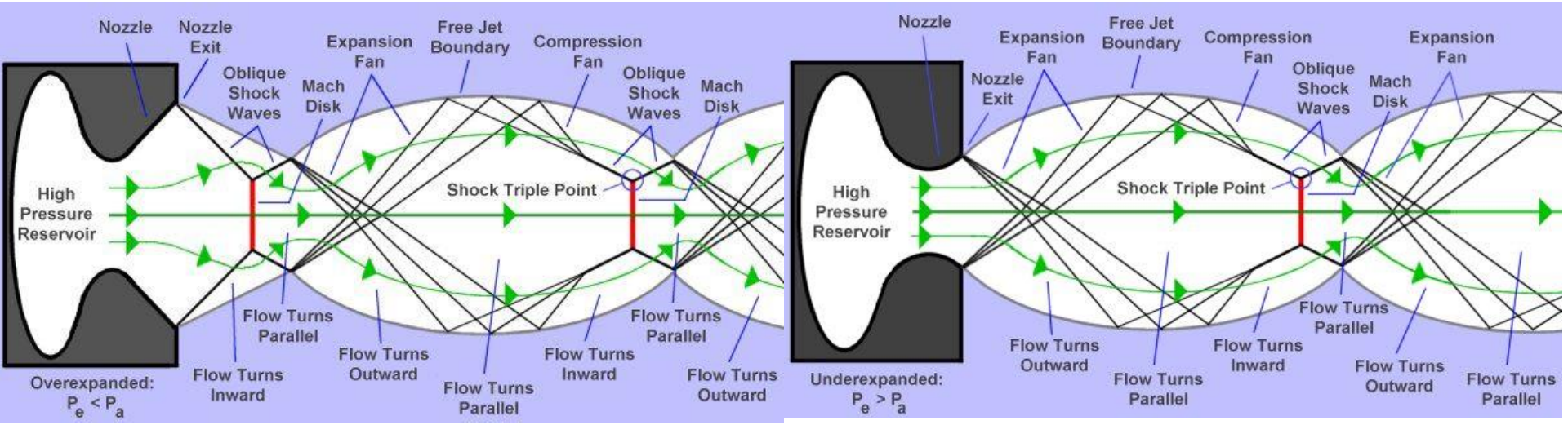
他们分别在什么情况下会出现？





§7.2可压缩层流运动

自学

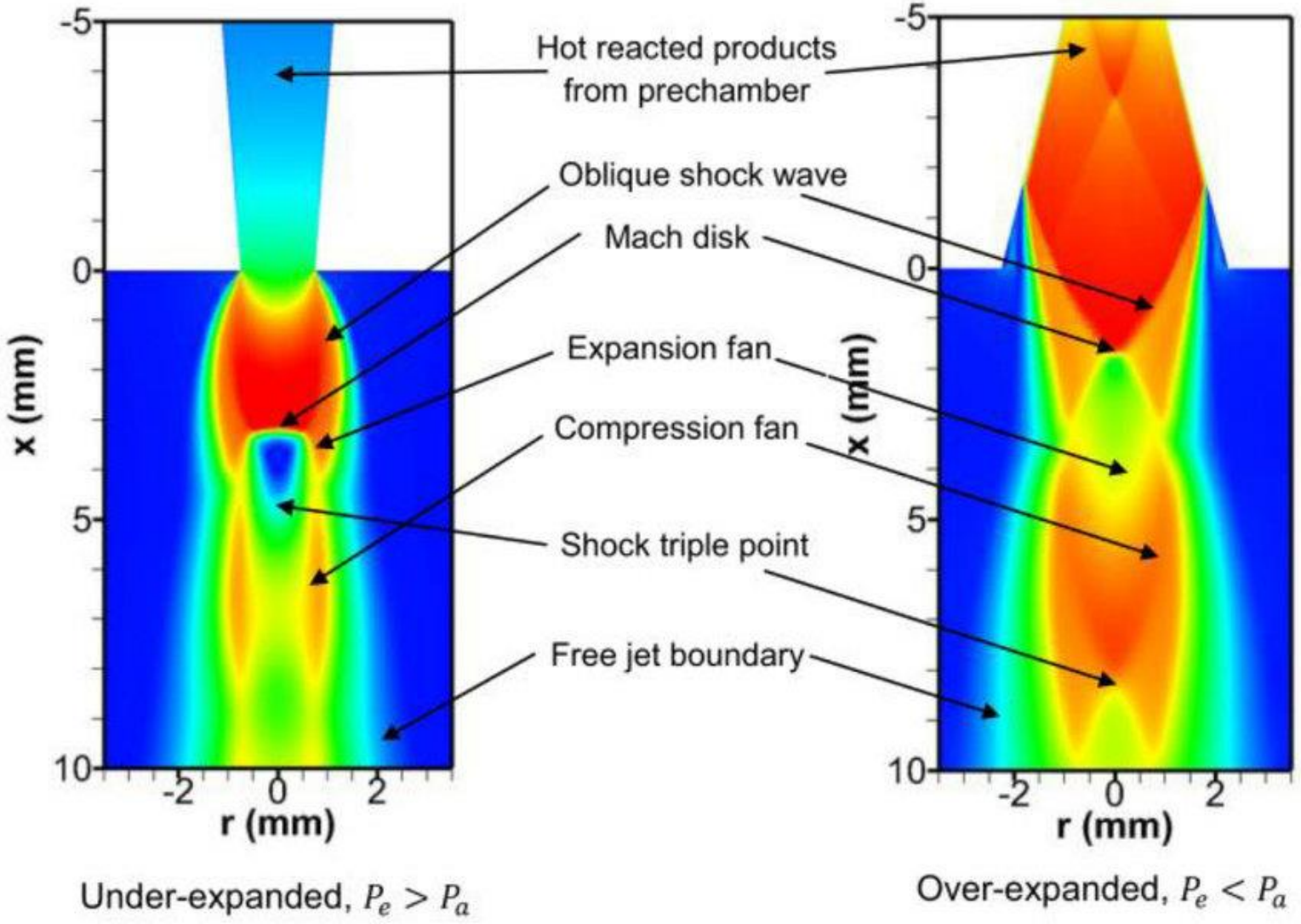


Over-expanded and **under-expanded** supersonic jet

If the pressure of the jet leaving the nozzle exit is still above ambient pressure, then a nozzle is said to be "**under-expanded**";
if the jet is below ambient pressure, then it is "**over-expanded**".

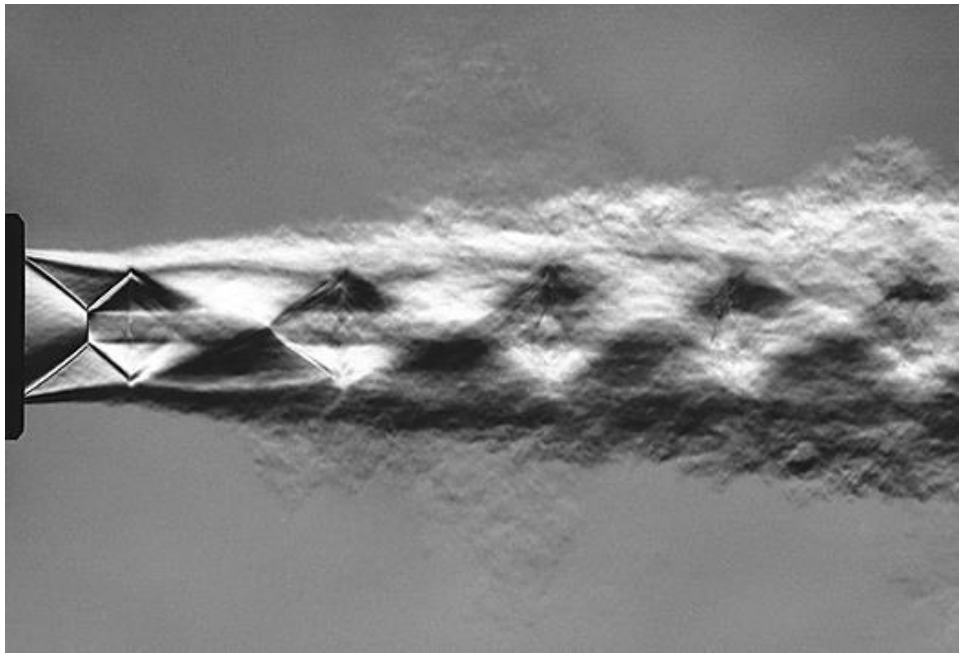
§7.2可压缩层流运动

自学

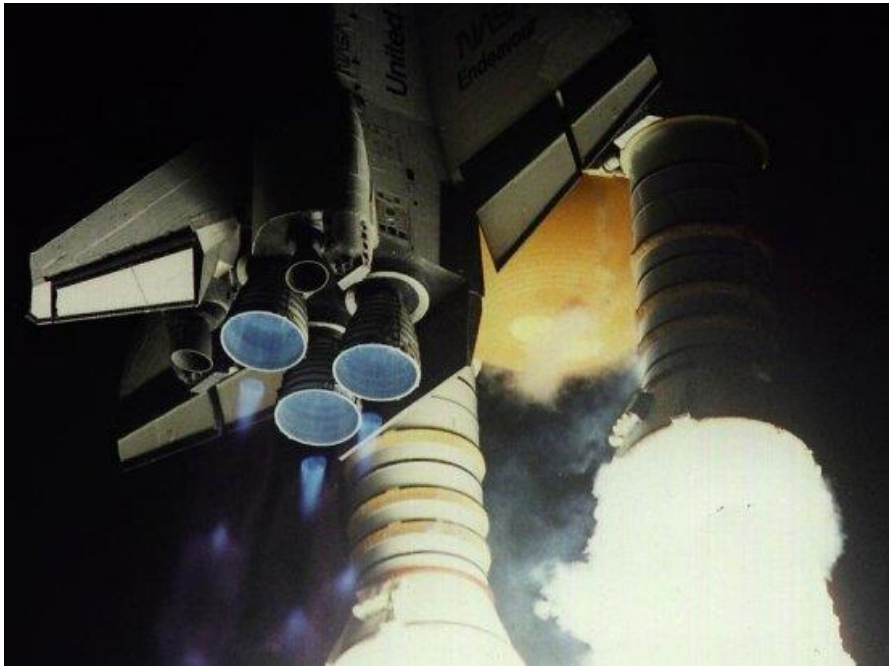


§7.2可压缩层流运动

自学





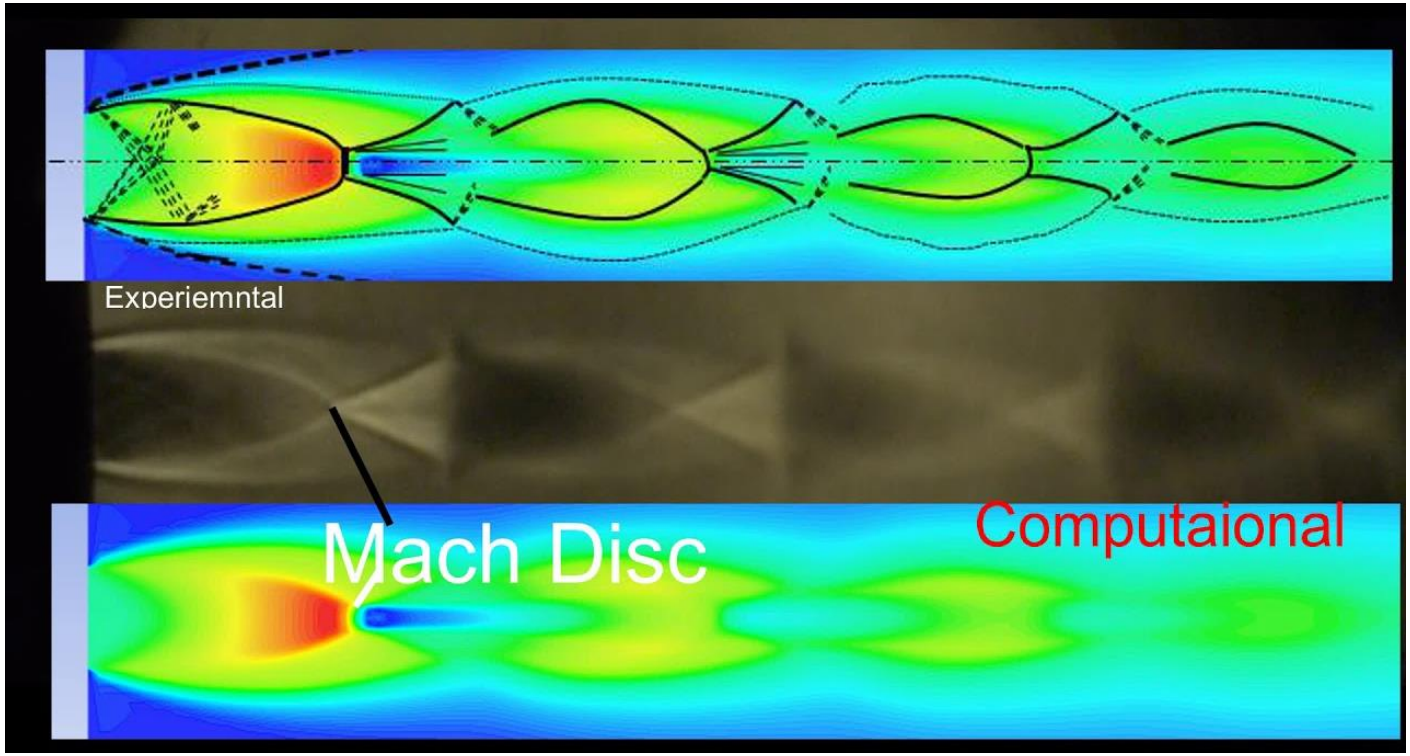


The complex interactions between shock waves and expansion waves in an "**over-expanded**" supersonic jet.

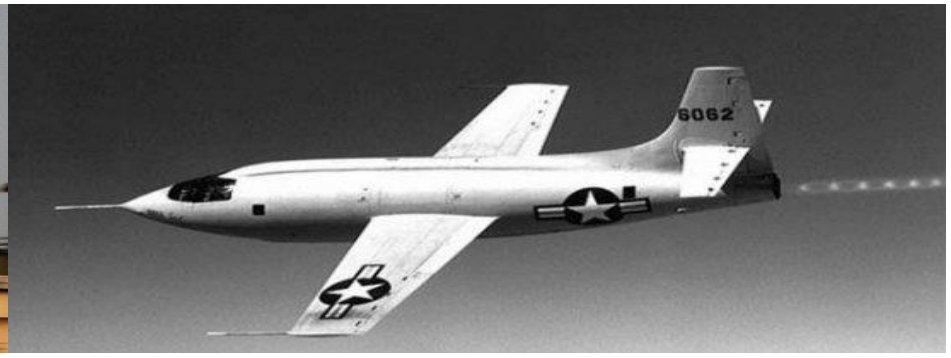
§7.2可压缩层流运动

自学

Under-expanded jet







§7.3可压缩湍流运动

可压缩湍流的研究主要在以下几个方面：

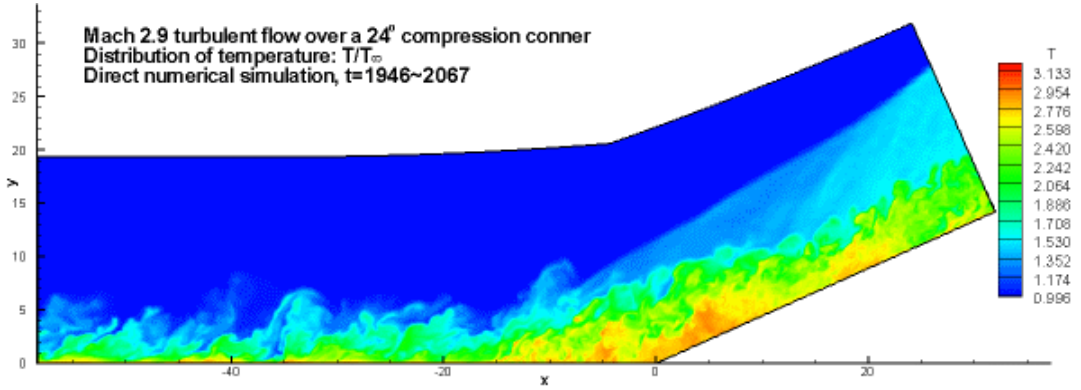
均匀各向同性湍流的统计性质和标度律、自由剪切湍流（射流、混合层）的可压缩效应、壁湍流的标度律和相干结构

由于可压缩湍流的复杂性，湍流的数值模拟研究还有很大空白。

下面主要介绍目前可压缩充分发展湍流的基本特点。

1. 可压缩效应在湍流中的表现：

（1）湍流场中的激波结构：大纵横比的结构

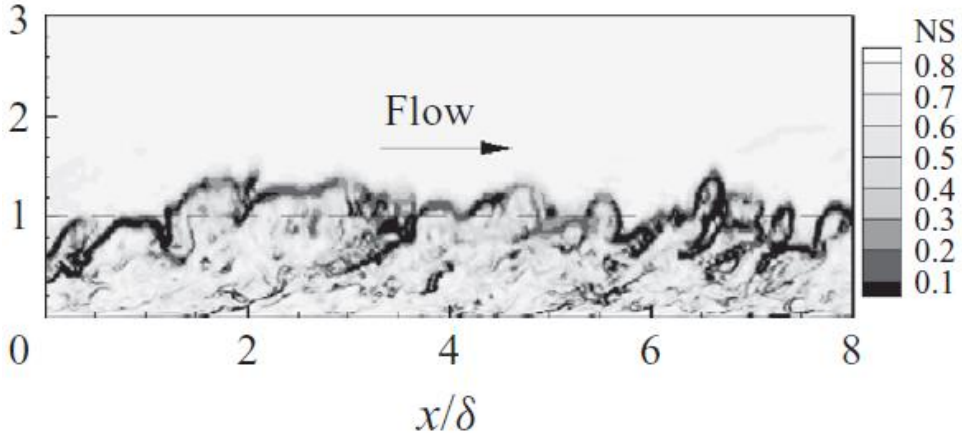


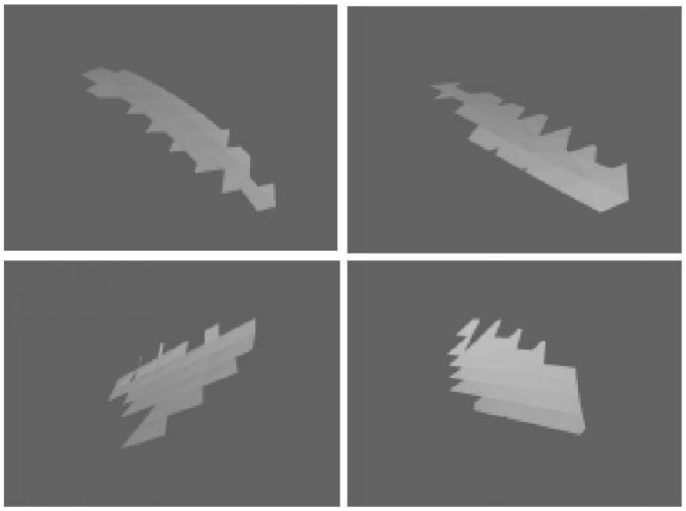
§7.3可压缩湍流运动

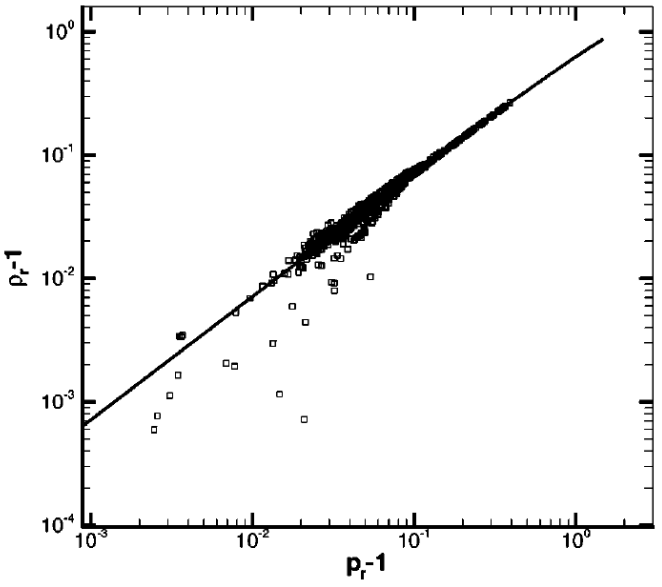
湍流场中有激波存在时，将产生局部强耗散区。

强压力梯度可导致壁面阻力和热流的增加。

来流马赫数较高时流场中会产生小激波（shocklets），在其上下游仍然满足R-H关系。







§7.3可压缩湍流运动

（2）扰动波及特征马赫数

Mach波：无粘流动压力扰动波仅沿特征方向向下游传播

湍流Mach数：$M_t = q_{rms}/a, q_{rms} = \sqrt{\left(\left\langle u'^2\right\rangle + \left\langle v'^2\right\rangle + \left\langle w'^2\right\rangle\right)}$

扰动Mach数：$M' = q'/a, q' = \sqrt{\left(u'^2 + v'^2 + w'^2\right)}$

梯度Mach数：$M_g = \dfrac{S^* L}{\bar{a}}, S^* = \left|\partial \bar{u}_i / \partial x_i\right|$

当$M_g > 1$时，在畸变过程中压力脉动没有足够的时间充分发展，其效应可忽略。

摩擦Mach数：$M_\tau = \dfrac{u_\tau}{\bar{a}}$

对流Mach数：$M_c = \dfrac{u_1 - u_2}{a_1 + a_2}$

§7.3可压缩湍流运动

（3）脉动速度的散度

$$\nabla \bullet u = -\frac{D\ln\rho}{Dt} = -\frac{1}{\gamma}\frac{D(\ln p)}{Dt} + \frac{1}{\gamma}\frac{D(\ln s)}{Dt}$$

在小扰动假设下 $\nabla \bullet u = \frac{1}{C_p}\frac{\partial s'}{\partial t} - \frac{1}{\gamma\rho_0}\frac{\partial p'}{\partial t} = -\frac{1}{\gamma p_0}\frac{\partial p'}{\partial t}$

脉动速度散度的产生仅确定于扰动压力随时间的变化。

（4）**Morkovin假定**：

(1) 当M_t较小时，湍流流场中的热力学量（密度、温度、压力）与平均量相比是小量，湍流的可压缩效应主要反映在对平均流的影响。

(2) 可以通过**某些变换**将平均密度变化考虑进去，使湍流的统计平均特征与不可压湍流相似。

§7.3可压缩湍流运动

2. 可压缩湍流的平均运动方程

雷诺分解：$f = \langle f \rangle + f'$

连续方程

$$\frac{\partial \langle \rho \rangle}{\partial t} + \frac{\partial \left(\langle \rho \rangle \langle u_j \rangle \right)}{\partial x_j} = -\frac{\partial \langle \rho' u_j' \rangle}{\partial x_j}$$

动量方程

$$\frac{\partial}{\partial t}\left(\langle \rho \rangle \langle u_i \rangle + \langle \rho' u_i' \rangle \right) + \frac{\partial}{\partial x_j}\left(\langle \rho \rangle \langle u_i \rangle \langle u_j \rangle + \langle \rho' u_i' u_j' \rangle \right)$$

$$= -\frac{\partial \langle p \rangle}{\partial x_i} + \frac{\partial}{\partial x_j}\left(-\langle \rho \rangle \langle u_i' u_j' \rangle - \langle u_j \rangle \langle \rho' u_i' \rangle - \langle u_i \rangle \langle \rho' u_j' \rangle \right) + \frac{\partial \langle \tau_{ij} \rangle}{\partial x_j}$$

能量方程

$$\frac{\partial}{\partial t}\left(\langle \rho \rangle \langle E \rangle + \langle \rho' E' \rangle \right) + \frac{\partial}{\partial x_j}\left(\langle \rho \rangle \langle u_j \rangle \langle E \rangle + \langle \rho' u_j' E' \rangle \right) = \frac{\partial}{\partial x_j}\left(\kappa \frac{\partial \langle T \rangle}{\partial x_j} \right) - \frac{\partial \langle p \rangle \langle u_j \rangle}{\partial x_j}$$

$$-\frac{\partial}{\partial x_j}\left(\langle \rho \rangle \langle u_j' E' \rangle + \langle E \rangle \langle \rho' u_j' \rangle + \langle u_j \rangle \langle \rho' E' \rangle \right) - \frac{\partial \langle p' u_j' \rangle}{\partial x_j} + \frac{\partial \langle u_i \rangle \langle \tau_{ij} \rangle}{\partial x_j} + \frac{\partial \langle u_i' \tau_{ij}' \rangle}{\partial x_j}$$

§7.3可压缩湍流运动

3. 密度加权平均(Favre平均)

密度加权平均量等于该量和密度乘积的平均值与平均密度之商。

密度加权平均量用上标 '一' 表示： $\bar{q}=\dfrac{\langle \rho q\rangle}{\langle \rho\rangle}$

加权平均的分解式为 $q=\bar{q}+q''$

性质：(1)加权平均量的系综平均等于原加权平均量 $\langle \bar{q}\rangle=\bar{q}$

(2)加权分解的脉动量的加权平均等于零 $\overline{q''}=0,\langle \rho q''\rangle=0$

(3)加权平均分解的脉动量的系综平均等于该量的系综平均与加权平均之差

$$\langle q''\rangle=\langle q\rangle-\bar{q}=-\langle \rho' q\rangle/\langle \rho\rangle=q''-q'$$

§7.3可压缩湍流运动

可压缩湍流的加权平均方程

u_i，θ，E，H 等采用加权平均分解，而 ρ，p，τ_{ij} 等采用系综平均分解，代入可压缩运动方程组，得：

$$\frac{\partial}{\partial t}\left(\langle\rho\rangle+\rho'\right)+\frac{\partial}{\partial x_i}\left(\rho\bar{u}_i+\rho u_i''\right)=0$$

$$\frac{\partial}{\partial t}\left(\rho\bar{u}_i+\rho u_i''\right)+\frac{\partial}{\partial x_j}\left(\rho\bar{u}_i\bar{u}_j+\rho u_i''\bar{u}_j+\rho u_j''\bar{u}_i+\rho u_i''u_j''\right)=-\frac{\partial}{\partial x_i}\left(\langle p\rangle+p'\right)+\frac{\partial}{\partial x_j}\left(\langle\tau_{ij}\rangle+\tau_{ij}'\right)$$

$$\frac{\partial}{\partial t}\left(\rho\bar{H}+\rho H''\right)+\frac{\partial}{\partial x_j}\left(\rho\bar{u}_j\bar{H}+\rho\bar{u}_jH''+\rho u_j''\bar{H}+\rho u_j''H''\right)$$

$$=\frac{\partial}{\partial t}\left(\langle p\rangle+p'\right)+\frac{\partial}{\partial x_j}\left(\bar{u}_i\langle\tau_{ij}\rangle+u_i''\langle\tau_{ij}\rangle+\bar{u}_i\tau_{ij}'+u_i''\tau_{ij}'\right)+\frac{\partial}{\partial x_j}\left(\kappa\frac{\partial T}{\partial x_j}\right)$$

§7.3可压缩湍流运动

对上述方程作系综平均，利用加权平均和系综平均的性质，整理可得

连续方程 $$\frac{\partial}{\partial t}\langle\rho\rangle+\frac{\partial}{\partial x_i}\left(\langle\rho\rangle\bar{u}_i\right)=0$$

动量方程 $$\frac{\partial}{\partial t}\left(\langle\rho\rangle\bar{u}_i\right)+\frac{\partial}{\partial x_j}\left(\langle\rho\rangle\bar{u}_i\bar{u}_j\right)=-\frac{\partial\langle p\rangle}{\partial x_i}+\frac{\partial}{\partial x_j}\left(\langle\tau_{ij}\rangle-\left\langle\rho u_i''u_j''\right\rangle\right)$$

能量方程 $$\frac{\partial}{\partial t}\left(\langle\rho\rangle\bar{H}\right)+\frac{\partial}{\partial x_j}\left(\langle\rho\rangle\bar{u}_j\bar{H}+\left\langle\rho u_j''H''\right\rangle\right)$$

$$=\frac{\partial\langle p\rangle}{\partial t}+\frac{\partial}{\partial x_j}\left(\bar{u}_i\langle\tau_{ij}\rangle+\langle u_i''\rangle\langle\tau_{ij}\rangle+\left\langle u_i''\tau_{ij}'\right\rangle\right)+\frac{\partial}{\partial x_j}\left(\langle\kappa\rangle\frac{\partial\bar{T}}{\partial x_j}\right)$$

$$\left(\because\quad\frac{\partial}{\partial x_j}\left\langle\kappa\frac{\partial T}{\partial x_j}\right\rangle\approx\frac{\partial}{\partial x_j}\left(\langle\kappa\rangle\frac{\partial\bar{T}}{\partial x_j}\right)\right)$$

§7.3可压缩湍流运动

4. 平板可压缩湍流边界层

来流马赫数小于5的绝热壁超声速湍流边界层特性和马赫数无关；即$Ma_\infty < 5$时，边界层特性和不可压湍流边界层相似。

在Morkovin假定下，超声速绝热壁边界层类似于低密度、高粘性的低雷诺数不可压流动，其湍流马赫数Ma_t<<1。

速度剖面：

粘性底层： $\mu\dfrac{\partial U}{\partial y}=\rho_w u_\tau^2 \qquad U_c^+=y^+$

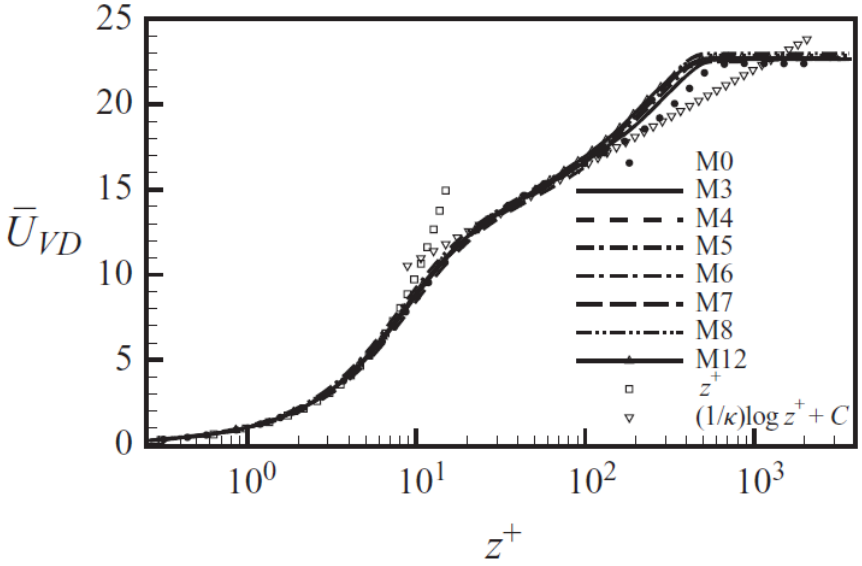
van Driest变换

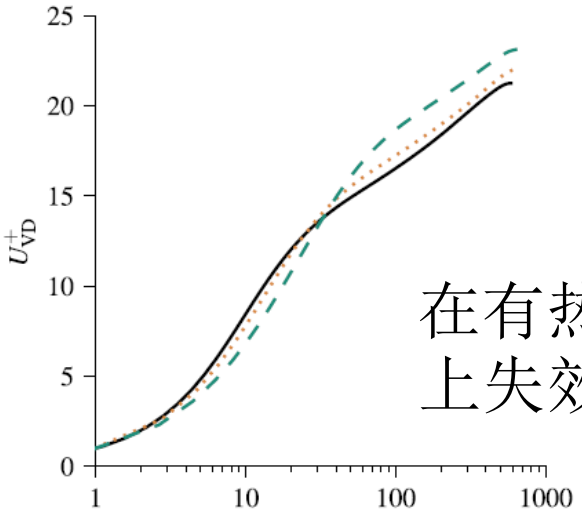
$$U_c=\int_0^U\left(\frac{T}{T_w}\right)^\alpha \mathrm{d}U=\int_0^U\left(\frac{\rho}{\rho_w}\right)^{-\alpha}\mathrm{d}U$$

对数区： $-\left\langle\rho u'v'\right\rangle=\rho_w u_\tau^2$ （等应力层）

$$\left(\frac{\rho}{\rho_w}\right)^{1/2}\frac{\partial U}{\partial y}=\frac{1}{\kappa}\frac{u_\tau}{y} \qquad U_c^+=\frac{1}{\kappa}\ln y^+ + B \qquad u_*=u_\tau\left(\rho_w/\rho\right)^{1/2}$$

§7.3可压缩湍流运动



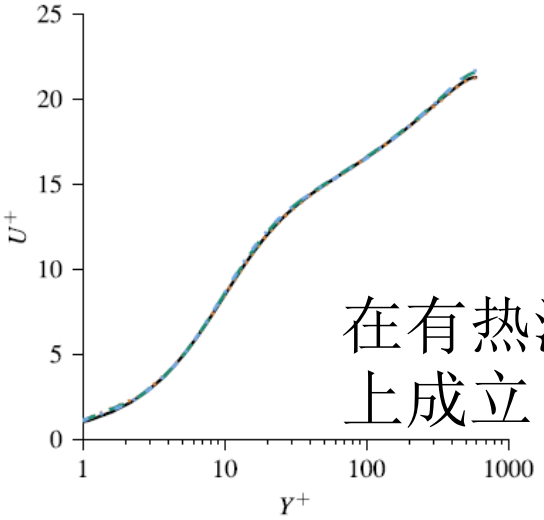


在有热流的壁面上失效！

基于等应力和对数律的变换(了解)

$$Y^+ = \frac{y\left(\tau_w \bar{\rho}\right)^{1/2}}{\bar{\mu}}$$

$$U^+ = \int_0^{u^+} \frac{\bar{\rho}}{\rho_w}\left[1 + \frac{1}{2\bar{\rho}}\frac{d\bar{\rho}}{dy}y - \frac{1}{\bar{\mu}}\frac{d\bar{\mu}}{dy}y\right]du^+$$



在有热流的壁面上成立！

Ref:Trettel, A., & Larsson, J. (2016). Mean velocity scaling for compressible wall turbulence with heat transfer. *Physics of Fluids*, *28*(2), 026102.

§7.3可压缩湍流运动

参考温度解法求壁面阻力系数：

令$\xi = x, \eta = \int_0^y \frac{\bar{\rho}}{\rho^*} dy$，且假定$-\overline{\rho u' v'} = -\overline{\rho^* u^* v^*}$，$\bar{\rho}\bar{\mu} = \rho^* \mu^*$，右上角带*的量表示相当于某一参考温度$\bar{T}^*$的量，可变换为与不可压缩湍流边界层一致的方程形式：

$$\frac{\partial u^*}{\partial \xi} + \frac{\partial v^*}{\partial \eta} = 0$$

$$\rho^* u^* \frac{\partial u^*}{\partial \xi} + \rho^* v^* \frac{\partial u^*}{\partial \eta} = \frac{\partial}{\partial \eta}\left(\mu^* \frac{\partial u^*}{\partial \eta} - \rho^* u'^* v'^* \right)$$

借助于不可压缩情形的解，可以求得：$C_f^* = \frac{2\tau_w}{\rho^* U_\infty^2} = \varphi\left(Re_\xi^*\right)$

§7.3可压缩湍流运动

$$C_f^* = \frac{2\tau_w}{\rho^* U_\infty^2} = \varphi\left(Re_\xi^*\right) \qquad Re_\xi^* = \frac{\rho^* U_\infty \xi}{\mu^*} = \frac{\rho^* \mu_\infty}{\rho_\infty \mu^*} \frac{\rho_\infty U_\infty x}{\mu_\infty} = \frac{\rho^* \mu_\infty}{\rho_\infty \mu^*} Re_x = \frac{T_\infty}{T^*} \frac{\mu_\infty}{\mu^*} Re_x$$

因此可得可压缩情形的壁面阻力系数：

$$C_f = \frac{\rho^*}{\rho_e} C_f^* = \frac{T_\infty}{T^*} C_f^* = \frac{T_\infty}{T^*} \varphi\left(Re_\xi^*\right) = \frac{T_\infty}{T^*} \varphi\left(\frac{T_\infty}{T^*} \frac{\mu_\infty}{\mu^*} Re_x\right)$$

其中

$$C_f^* = 0.026 Re_\xi^{*-1/7} \left(\frac{T_\infty}{T^*} \frac{\mu_\infty}{\mu^*}\right)^{1/7} \qquad C_f = 0.026 Re_x^{-1/7} \left(\frac{T^*}{T_\infty}\right)^{-6/7} \left(\frac{\mu^*}{\mu_\infty}\right)^{1/7}$$

根据粘性系数的幂次律

$$\mu = \mu^* \left(\frac{T}{T^*}\right)^n$$

$$\frac{C_f}{C_{fi}} = \left(\frac{T^*}{T_\infty}\right)^{\frac{n-6}{7}}$$

§7.3可压缩湍流运动

对于空气：$\gamma = 1.4, Pr = 0.72, n = 0.76$

$$\frac{C_f}{C_{fi}} = \left(\frac{T^*}{T_\infty}\right)^{\frac{n-6}{7}}$$

$$C_{fi} = 0.026 Re_x^{-1/7}$$

$$T_{aw} = T_\infty \left(1 + \frac{\gamma - 1}{2} Pr^{1/3} M_\infty^2\right) \approx T_\infty \left(1 + 0.1793 M_\infty^2\right)$$

由Crocco-Busemann关系：

$$\frac{T^*}{T_\infty} = 0.5 + 0.5\frac{T_w}{T_\infty} + 0.0299 M_\infty^2$$

$$C_f \approx 0.026 Re_x^{-1/7} \left(0.5 + 0.5\frac{T_w}{T_\infty} + 0.0299 M_\infty^2\right)^{-0.75}$$

§7.3可压缩湍流运动

$$C_f \approx 0.026 Re_x^{-1/7} \left(0.5 + 0.05 \frac{T_w}{T_\infty} + 0.0299 M_\infty^2 \right)^{-0.75}$$

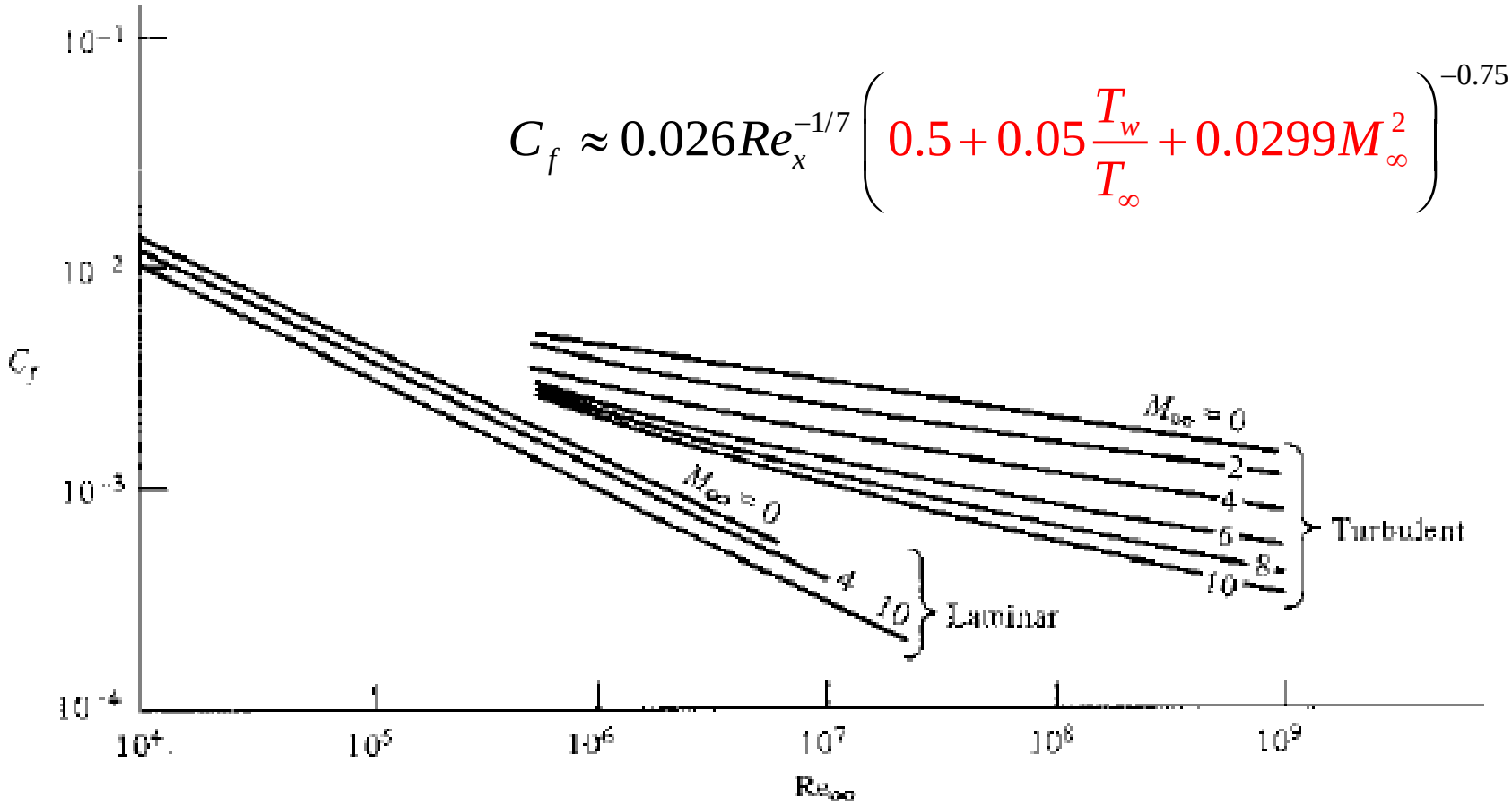


Figure 19.1 Turbulent friction drag coefficient for a flat plate as a function of Reynolds and Mach numbers. Adiabatic wall Pr = 0.75. For contrast, some laminar results are shown. (*Data are from the calculations of van Driest, Reference 47.*)

§7.3可压缩湍流运动

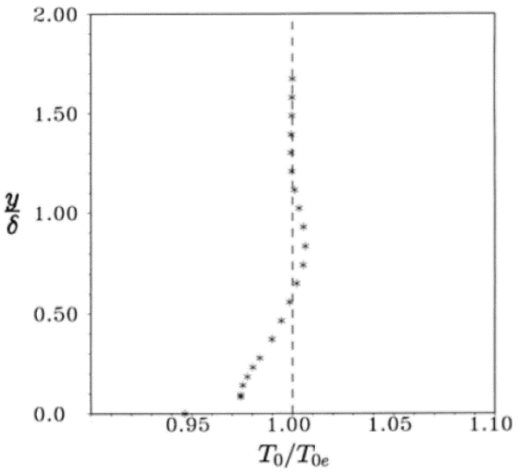
5. 强雷诺比拟(SRA)

超声速湍流边界层中，温度脉动和速度脉动的相关系数为-1.

$$\frac{T_0'}{T_0}=\frac{T'+Uu'/c_p}{T+U^2/2c_p}\ll 1$$

$$T'\approx -Uu'/c_p$$

$$R_{T'u'}\approx -1$$



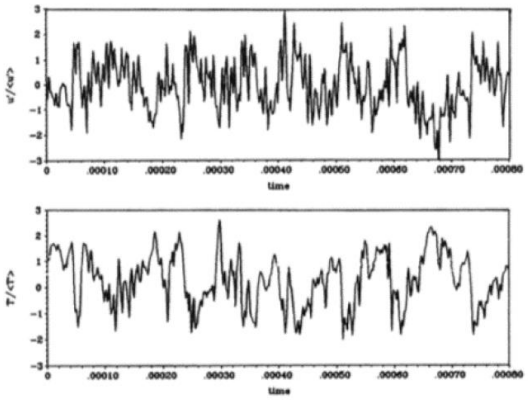


图 6-11 Ma=2.9 的超声速湍流边界层中 y/d=0.6 处速度脉动（上图）和温度脉动（下图）的时间序列

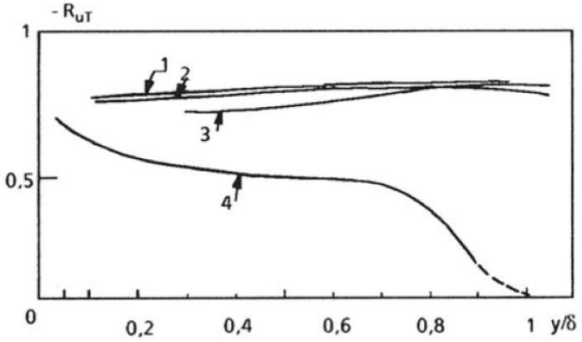


图 6-12 超声速湍流边界层中温度速度相关。曲线 1：Me=2.32, Reθ=5650；曲线 2,3：Me=:1.73, Reθ=5700；曲线 4：Me<<1, Reθ=5000

§7.3可压缩湍流运动

湍流普朗特数

根据涡粘系数和涡扩散系数的定义：

$$-\overline{u'v'}=\nu_t\frac{\partial U}{\partial y},\qquad -\overline{T'v'}=\kappa_t\frac{\partial T}{\partial y}$$

湍流普朗特数：

$$Pr_t=\frac{\nu_t}{\kappa_t}=\frac{\overline{u'v'}}{\overline{T'v'}}\frac{\partial T/\partial y}{\partial U/\partial y}$$

由强雷诺比拟：

$$T_0=T+U^2/2c_p\approx \text{const.}$$

$$T_0'=T'+Uu'/c_p\approx 0$$

$$\Rightarrow\quad \frac{\partial T}{\partial y}=-\frac{U}{c_p}\frac{\partial U}{\partial y}$$

$$-\overline{T'v'}=\frac{U}{c_p}\overline{u'v'}$$

因此，$Pr_t=1$

超声速绝热壁边界层(Ma<5)中，热通量和动量通量具有相似性。

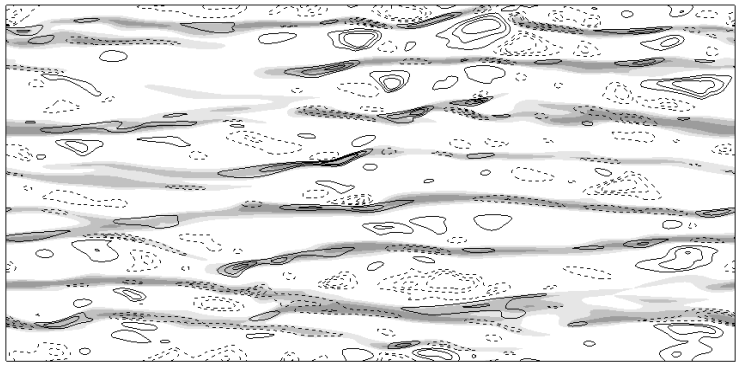
§7.3可压缩湍流运动

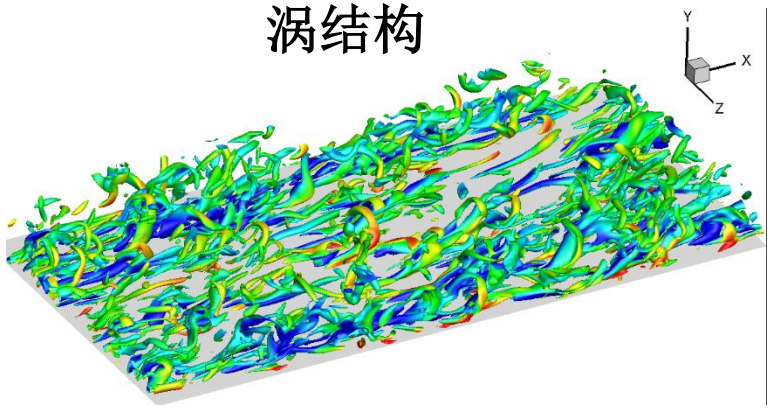
6. 相干结构：与不可压缩壁湍流类似

低速条带

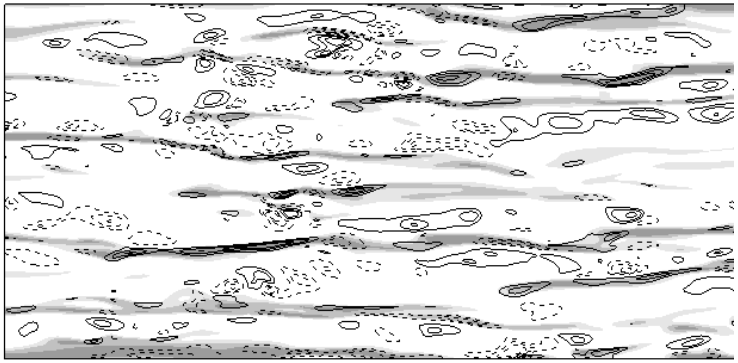
涡结构

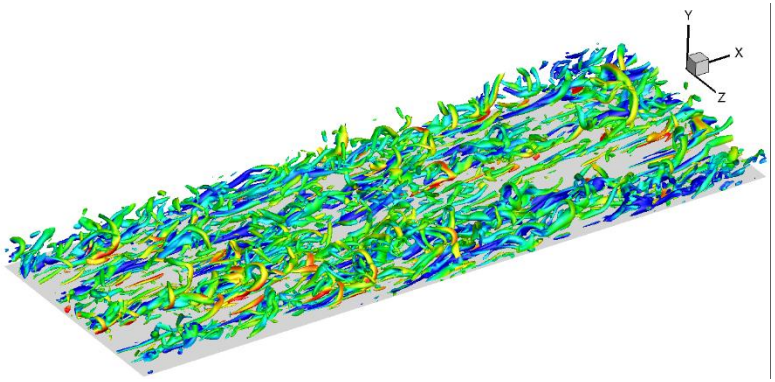
Ma=1.5
Re=3000





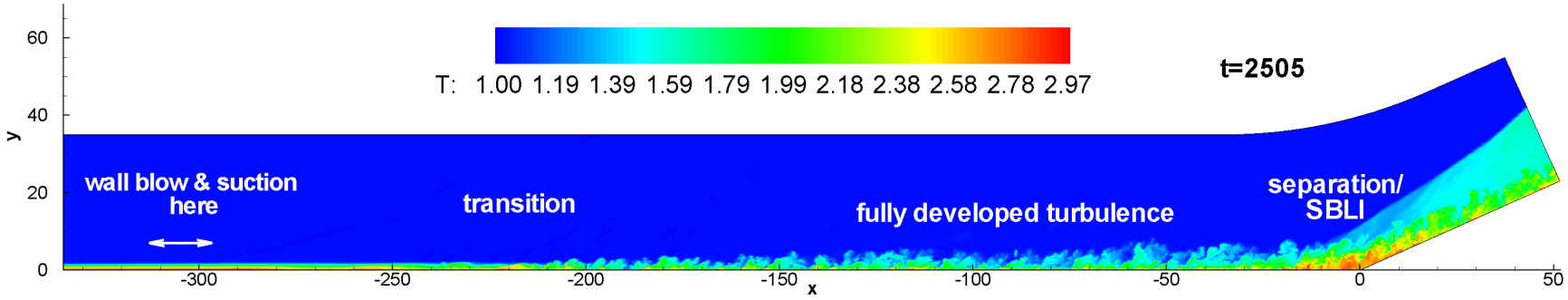
Ma=3.0
Re=4880



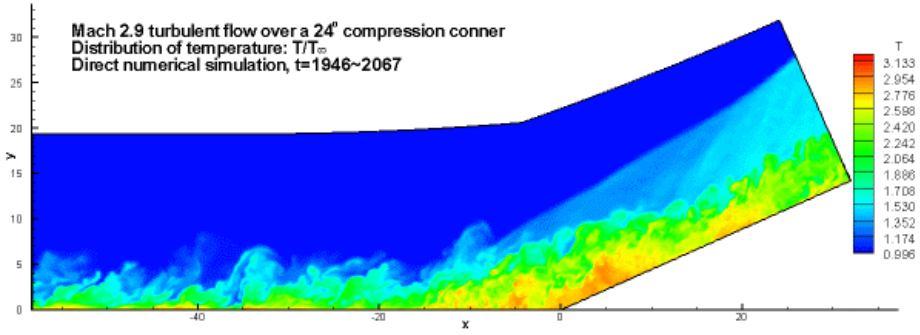


§7.3可压缩湍流运动

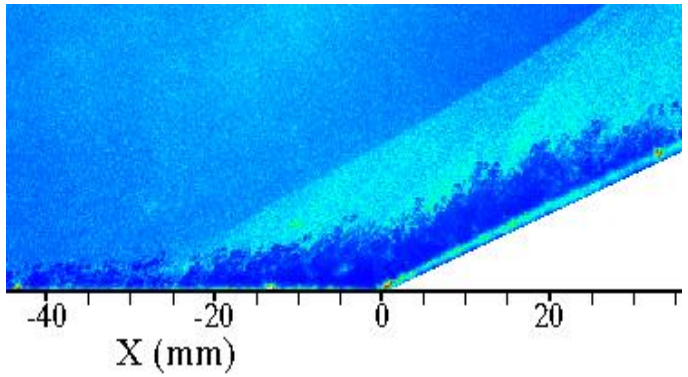
激波–边界层干扰



压缩折角激波-边界层干扰DNS



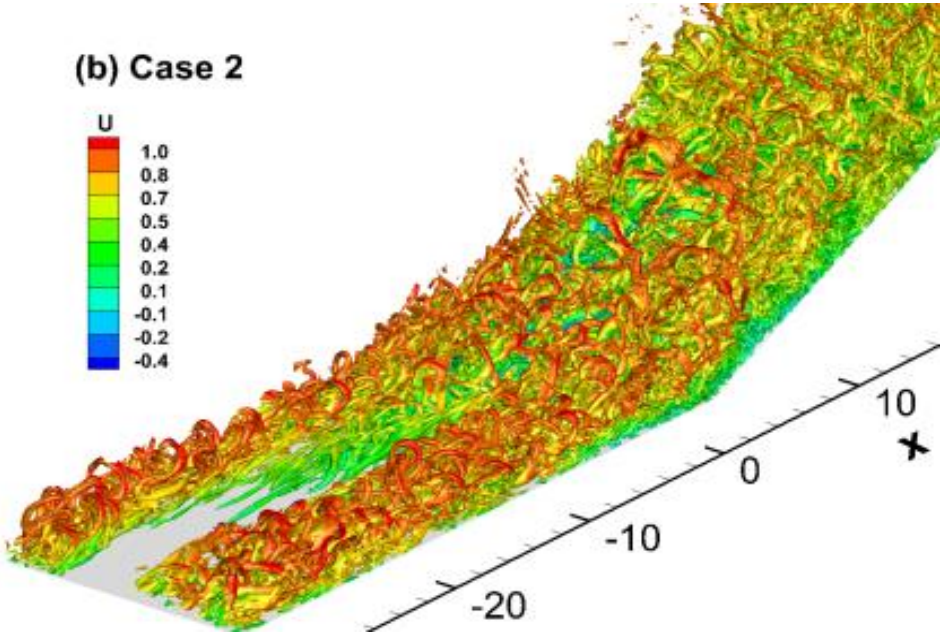
DNS：中截面上的瞬时温度

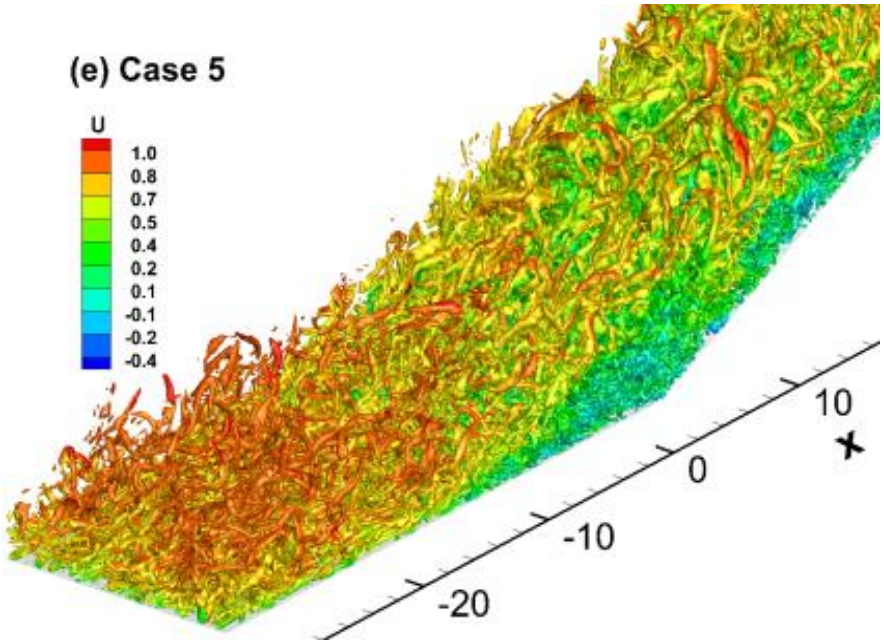


PIV 实验 （易仕和）

§7.3可压缩湍流运动

激波-边界层干扰





转捩早期： 强三维结构

充分发展湍流：相对均匀

傅德薰、马延文、李新亮、王强， 可压缩湍流直接数值模拟， 科学出版社

OpenCFD-SC, OpenCFD-EC 代码已开源！

本章小结

本章主要介绍了不可压缩温度边界层和可压缩边界层。要点如下：

1. **边界层的厚度估计**
2. **边界层方程、边界条件和变换**
3. **层流边界层的相似性解**
4. **湍流边界层**

无量纲参数的物理含义

由于可压缩边界层的复杂性，在此仅介绍了最基本的内容。有兴趣的同学请参考相关教材。

课程内容结束！谢谢大家！

祝各位考试顺利！